# MoCHII: Physics, Atomic Data, Algorithms, and Validation


**Abstract**

MoCHII (MOnte Carlo for H II regions) adds a photoionization gas layer to the validated octree-AMR dust radiative-transfer engine of MoCafe v2.00. This memo documents the ionizing frequency band and its zero-variance transport, the complete atomic-data inventory with sources and the analytic fitting functions evaluated at run time, the iteration and equilibrium algorithms (ionization, thermal balance, trace-metal cascade, explicit diffuse field), the $n$-level emissivity solver, the ported nebular continuum and Storey & Hummer recombination lines, and the quantitative validation gates: rate integrals vs. analytic attenuation (bias +0.004%); the Strömgren sphere vs. an exact 1D reference (+0.56% in ionized volume; AMR $\equiv$ uniform); the thermal H/He nebula (median 0.14% in $T_e$); the MOCASSIN Lexington benchmarks ($T_e$ = 8211 K at HII40 and 7025 K at HII20 with the density-suppressed cooling default, matching a current CLOUDY c25.00 run to 0.7 K and 1.7%; the zero-density limit gives 7894 and 6376 K, and both modern codes sit above the 2000-era published ranges); and the line-flux comparison (H$\beta$ within a few percent of the published values, the optical collisional lines within the published spread). Also documented: the solution-driven I-front re-refinement (re-refined vs. native high-resolution octree +0.008% in ionized volume with 4.5$\times$ fewer cells) and an Illustris-TNG post-processing demonstration; the completed dust coupling (ionizing-band dust absorption with the dusty Strömgren reduction 0.724 reproduced to +0.34%; EUV/FUV dust *scattering* with forced-first interaction sampling, validated by bracketing; live ionization-dependent dust with a PAH-specific survival factor; and the equilibrium dust temperature + IR spectrum consuming the transported grain heating, energy closure 1.000); and the incremental metal additions (argon, magnesium, iron, and silicon/chlorine/calcium) as pure data operations.


# Contents

# 1. Scope and design rules

MoCHII copies validated modules from MoCafe v2.00 (octree, event-driven traversal, MPI shared memory, HDF5/FITS I/O) and adds gas microphysics under three design rules: (1) *atomic data are fitted offline, evaluated online* — the Python pipeline under `tools/fitting/` is the auditable source of every coefficient in `data/atomic/`, and Fortran only evaluates closed forms; (2) *adding an ion is a data operation*, realized by the species registry (Section 4.5); (3) *validation by quantitative gates*, each capability compared against an analytic or external reference.

The capabilities documented here: the ionizing band + rate integrals; ionization equilibrium + opacity feedback (case B, fixed $T_e$); thermal balance + trace-metal frame; the explicit diffuse field (case A); solution-driven I-front re-refinement + snapshot post-processing; the dust coupling (absorption competition, scattering, live PAH split, grain heating/temperature, and the SEDust stochastic-emission coupling); the incremental ions (argon, magnesium, iron, silicon, chlorine, calcium); plus the output-time diagnostic layer: $n$-level emissivities, Storey & Hummer recombination lines, and the nebular continuum.

# 2. The ionizing band and transport

## 2.1 Frequency bins and the source

The ionizing band covers photon energies $[E_{\rm min}, E_{\rm max}] = [13.598, 100]$ eV with $n_\nu$ log-spaced bins (`par%nnu_ion`; gates use 32). Bin centers are geometric means; the source spectrum (Planck $B_\nu(T_\star)$ or a 2-column file) is integrated on a 32-point sub-grid within each bin and normalized so that $\sum_b L_b = L_{\rm band}$ (`par%luminosity`). Packets sample the bin from the luminosity CDF and carry $L_{\rm pkt} = L_{\rm band}/N$. Bin-count convergence of the rate integrals (measured in the rate-integral gate): 16 bins $\to$ 2.2%, 32 $\to$ 2.1%, 64 $\to$ 0.02% against a 2048-bin reference.

**Threshold-aligned bins (default).** With `par%ion_align_edges` (default on) the ionizing-band bin *edges* are pinned onto the ionization thresholds — H I (13.6 eV), He I (24.6 eV), He II (54.4 eV), plus the active metal photoionization thresholds gathered from the registry — so that no bin straddles a threshold. The sub-segments between consecutive thresholds are log-uniform, and bins are distributed among the segments by log width (largest-remainder, floor 1). When the thresholds outnumber the bins the lowest-priority (metal) edges are dropped gracefully — the H/He edges are always kept — so the option is safe at any `nnu_ion`. Setting it `.false.` recovers the legacy pure-log grid, bit-identical to the earlier behavior. The alignment removes a real discretization error: a coarse 32-bin log grid straddles the 24.6 eV He I edge and delivers $\sim$18% too many He-ionizing photons against a true 40 kK blackbody ($Q_{\rm HeI}/Q_{\rm HI} = 0.127$ versus the exact 0.108), over-ionizing He (and, downstream, the metals); aligning the edges restores the exact ratio at fixed bin count (Section 5.4).

**FUV extension.** With `par%add_fuv` the band grows a second, lower segment: `par%nnu_fuv` log bins on $[E_{\rm FUV}, E_{\rm min}]$ (`par%efuv_min`, default 6 eV) *below* the unchanged ionizing bins, so widening the band does not coarsen the ionizing resolution (the earlier single-segment caveat). `par%luminosity` keeps its meaning as the ionizing $[E_{\rm min}, E_{\rm max}]$ luminosity; the FUV part of the same source spectrum rides on top ($L_{\rm FUV}/L_{\rm ion} = 1.097$ for a $4\times10^4$ K Planck source at 6 eV), so the ionizing photon budget $Q_{\rm H}$ is preserved without manual band-ratio rescaling and packets carry $L_{\rm pkt} = (L_{\rm ion} + L_{\rm FUV})/N$. H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages (next subsection) — they penetrate the neutral gas beyond the I-front and photoionize Mg I (7.65 eV), C I (11.26 eV), S I (10.36 eV), and Fe I (7.90 eV) there, producing the PDR-side Mg II and C II that the ionizing-only band cannot.

## 2.2 Source components: multiple point sources and an external field

The band is fed by any number of *source components*: `par%nsource` internal point sources (0–16; positions `par%src_x/y/z`, ionizing luminosities `par%src_lum` — all set, or none for an equal split of `par%luminosity`, which becomes the sum) plus, independently, one isotropic external radiation field, on when `par%ext_intensity` $> 0$. Each packet draws a component in proportion to its band-total luminosity, is emitted from it, and draws its frequency bin from that component's own spectrum CDF; every packet carries $L_{\rm pkt} = \sum_c L_c/N$. A single-component run takes the legacy code path (the RNG stream is unchanged, so earlier results reproduce). The spectrum of internal source $i$ resolves as `par%src_spectrum_file` column $i$ > `src_tstar`($i$) Planck > the global `ion_spectrum` file > the global `tstar`; `src_spectrum_file` is one multi-column text file — column 1 the photon energy $E$ [eV] ascending, then one $L_E$ column for each source in arbitrary units, each independently normalized so source $i$'s ionizing segment carries `src_lum`($i$).

The external field is an isotropic band-integrated mean intensity $J$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] entering through an illuminating surface (`par%ext_geometry`): `'rec'`, the box faces (face-area-weighted — correct for a non-cubic box), or `'sph'`, the bounding sphere of radius `par%rmax` about the box center. The inward flux through the surface is the hemisphere integral of $J\cos\theta$, so the entering band power is

$$F_{\rm in} = \int_{2\pi} J\cos\theta\, d\Omega = \pi J, \qquad L = \pi J A_{\rm surf}. \tag{1}$$

Entry directions are cosine-weighted (Lambert) into the inward hemisphere, $\mu = \sqrt{\xi}$, which reproduces an isotropic interior field with energy density $u = 4\pi J/c$: an optically thin box tallies an interior mean intensity equal to $J$ (the gate of Section 5.8). Its spectrum resolves as `ext_spectrum` (file) > `ext_tstar` Planck > the global spectrum.

**Physical-unit spectra.** Every file slot is read in the column units set by `par%spectrum_type` (default `'shape'`, arbitrary units renormalized to the scale — the legacy behavior). The physical types `'per_ev'`, `'per_hz'`, `'per_ang'`, `'per_um'` give the first column as $E$ [eV], $\nu$ [Hz], or $\lambda$ [Å or $\mu$m] and the value columns as a luminosity density $L_X$ (internal sources) or an intensity density $J_X$ (external field) in the matching spectral unit; wavelength tables are converted to energy and sorted. The `'shape'` value columns are read as the per-eV density $L_E$ on the $E$ [eV] grid; a shape tabulated in wavelength or frequency must use the matching `per_*` type, whose conversion applies the $d\lambda/dE$ (or $d\nu/dE$) Jacobian, so a raw re-grid onto $E$ under `'shape'` would be wrong by $E^{-2}$. A physical spectrum is *absolute*: its bin luminosities follow the tabulated integral directly. If the scale (`luminosity` / `src_lum`($i$) / `ext_intensity`) is unset it is derived from the file and reported; if set, the spectrum is rescaled to it and acts as a shape. With `add_fuv` the same table fills the FUV bins with no separate normalization; a spectrum with no luminosity inside the band aborts.

**ISRF presets.** `par%ext_spectrum` may instead name one of three analytic interstellar radiation fields: `'draine'` (Draine 1978), `'habing'` (the Draine shape at Habing 1968 normalization, /1.71), and `'mathis'` (Mathis, Mezger & Panagia 1983), the shape of the SEDust radiation-field module. They are FUV-only (5–13.6 eV, exactly zero in the ionizing bins), so they require `add_fuv`, and `par%ext_scale` multiplies them. The Draine field is the energy-weighted photon polynomial $F_{\rm ph}(E) = 1.658\times10^{6}\,E^{2} - 2.152\times10^{5}\,E^{3} + 6.919\times10^{3}\,E^{4}$ with $J_E = e\,F_{\rm ph}$ ($e = 1.602\times10^{-12}$ erg eV$^{-1}$; $F_{\rm ph} = E\,N(E)$ is energy-weighted — a common pitfall). The presets carry the canonical FUV energy densities $u_{\rm FUV} = 8.94$, 5.23, $6.01\times10^{-14}$ erg cm$^{-3}$ (Draine / Habing / Mathis; $G_0 = 1.69$, 0.99, 1.14 in Habing units). When `ext_intensity` is set the preset or physical file is rescaled to that band-integrated $J$ (a note is printed; `ext_scale` is then ignored).

### 2.3 Opacity and the zero-variance edge walk

Each leaf carries the band absorption coefficient per code length

$$\kappa_b \;=\; n_{\rm H}\,[x_{\rm HI}\,\sigma_{\rm HI}(E_b) + y_{\rm He}\,(x_{\rm HeI}\sigma_{\rm HeI} + x_{\rm HeII}\sigma_{\rm HeII})(E_b)]\,d_{\rm cm}\;(+\,\kappa_{\rm dust}), \tag{2}$$

stored as the full $(n_\nu, n_{\rm leaf})$ array `kap_ion` — the grey $s_{\rm ext}$ rescaling of the dust SED band cannot apply because the gas opacity varies with both leaf and bin. With `par%ion_metal_abs` (default on when `use_metals`) the registry metals join the absorption, $\Delta\kappa_b = n_{\rm H}\sum_{\rm el} A_{\rm el}\sum_i f_i\,\sigma_{{\rm VFKY96},i}(E_b)\,d_{\rm cm}$, with the stage fractions $f_i$ from the same product chain as the cooling, iterated by the opacity feedback exactly like $x_{\rm HI}$. Above 13.6 eV this is a $\sim10^{-22}$ cm$^2$ per H perturbation next to H/He and dust; in the FUV bins it is the only *gas* opacity. The absorbed share returns to the metals through their own $\Gamma$ integrals, so the bookkeeping stays consistent (metal photoheating is not added to the thermal balance — negligible at trace abundances).

Where the band has no scattering and the diffuse field is emitted explicitly, every packet's contribution to the mean-intensity tally is computed *analytically* along its ray (the forced-first-flight estimator of MoCafe, made exact):

$$\Delta J_b(\text{leaf}) \;=\; w\,L_{\rm pkt}\,\frac{e^{-\tau_{\rm in}} - e^{-\tau_{\rm out}}}{\kappa_b}, \tag{3}$$

accumulated leaf by leaf until the domain edge (`raytrace_ion_to_edge_amr`). No interaction sampling is needed; the only Monte Carlo variance is in direction and bin sampling. The rate-integral gate verifies this estimator to a mean bias of +0.004%.

### 2.4 Rate integrals

With $V$ the leaf volume and $J_b$ the reduced tally, the code forms for each absorber $i$

$$\Gamma_i = \sum_b \frac{[4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)}{h\nu_b}, \qquad \mathcal{H}_i = \sum_b [4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)\left(1 - \frac{E_{{\rm th},i}}{E_b}\right), \tag{4}$$

the photoionization rate [s$^{-1}$] and photoheating [erg s$^{-1}$] for each particle of species $i$.

## 3. Atomic data

Every coefficient file in `data/atomic/` carries a provenance header (source, version, fit range, maximum fit error). The generators live in `tools/fitting/`.

## 3.1 Photoionization cross sections

The one-electron ions H I ($Z$=1) and He II ($Z$=2) use the *exact* ground-state hydrogenic cross section (Osterbrock & Ferland 2006, Eq. 2.4; `sigma_hydrogenic`, the same form as EXHALE):

$$\varepsilon = \sqrt{E/E_{\rm th}-1}, \qquad \sigma(E) = \frac{\sigma_0}{Z^2}\left(\frac{E_{\rm th}}{E}\right)^4 \frac{\exp\left(4-4\arctan\varepsilon/\varepsilon\right)}{1-e^{-2\pi/\varepsilon}}, \tag{5}$$

with $\sigma_0 = 6.30\times10^{-18}\,\mathrm{cm}^2$ the threshold value and $E_{\rm th}$ the true ionization potential (13.598 and 54.416 eV), tied to the physical edge (not $13.6Z^2$) so it stays aligned with the band grid. He I (two electrons) and every metal ion use the full VFKY96 single-shell analytic form (Verner, Ferland, Korista & Yakovlev 1996, ApJ 465, 487, Eq. 1):

$$x = \frac{E}{E_0} - y_0, \quad z = \sqrt{x^2+y_1^2}, \quad Q = 5.5 - \tfrac{1}{2}P, \quad \sigma(E) = \sigma_0\left[(x-1)^2+y_w^2\right] z^{-Q}\left(1+\sqrt{z/y_a}\right)^{-P}, \tag{6}$$

zero below threshold, $\sigma_0$ in Mb. Parameters were parsed directly from the authors' `phfit2.f` `PH2` (outer-shell) table and verified against the EXHALE implementation; H/He values are compiled in `photo_xsec.f90`, the registry ions in `element_<el>.txt`. For the elements *absent* from the VFKY96 outer-shell table (P, Cl, K — Opacity-Project gaps; the MOCASSIN `ph2.dat` read loop skips the same $Z$), the Verner & Yakovlev (1995) outer-shell subshell fit is used instead (`sigma_vy95`; `PHOTO2` lines): $\sigma = \sigma_0[(y-1)^2+y_w^2]y^{-(5.5+l-P/2)}(1+\sqrt{y/y_a})^{-P}$ with $y = E/E_0$ and $l$ the subshell orbital quantum number — the same VFKY96/VY95 hybrid as MOCASSIN `phFitEl`. Chlorine is the first user of this path. Among H/He, only the He I ground state uses VFKY96:

| ion | $E_{\rm th}$[eV] | $E_0$ | $\sigma_0$[Mb] | $y_a$ | $P$ | $y_w$ | $y_0$ | $y_1$ |
|---|---|---|---|---|---|---|---|---|
| He I | 24.587 | 13.61 | 949.2 | 1.469 | 3.188 | 2.039 | 0.4434 | 2.136 |

plus C I–III, N I–II, O I–III, Ne I–III, S I–IV, Ar I–IV (20 rows, in the element files). The rate-integral gate cross-checks the curves against EXHALE's independent implementations. The exact H I and He II cross sections give the threshold values $\sigma_0/Z^2 = 6.30$ and 1.575 Mb; the VFKY96 fits they replace for H/He agreed with the exact curve to 0.73% (H I) and 0.43% (He II), and $\sigma_{\rm HeI} = 7.43$ Mb (VFKY96).

## 3.2 H/He recombination

The recombination coefficients in the ionization balance are chosen by `par%recomb_model`. The default `'badnell_mao'` takes the total (case A) radiative recombination $\alpha_A$ from the same Badnell (2023) RR fit used for the metals [Eq. (11)], the ground-level ($n$=1) direct-capture rate $\alpha_1$ from the level-resolved parameterization of Mao & Kaastra (2016, A&A 587, A84), and forms case B by removing the ground term, $\alpha_B = \alpha_A - \alpha_1$. The Mao & Kaastra level fit is ($T_{\rm eV} = k_B T$; $a_0$ in $10^{-10}\,\mathrm{cm^3\,s^{-1}}$)

$$\alpha_1(T) = a_0\,10^{-10}\,T_{\rm eV}^{-b_0-c_0\ln T_{\rm eV}}\,\frac{1+a_2\,T_{\rm eV}^{-b_2}}{1+a_1\,T_{\rm eV}^{-b_1}}; \tag{7}$$

for the H-like ions (H II→H I, He III→He II) it is derived from the same exact hydrogenic cross sections as Eq. (5) under detailed balance, so $\alpha_1$ is exact. He II→He I adds the Badnell three-term dielectronic sum (negligible below $\sim 5\times10^4$ K); H II and He III are bare nuclei with no DR. The coefficients at $10^4$ K ($10^{-13}\,\mathrm{cm^3\,s^{-1}}$) are

| reaction | $\alpha_A$ | $\alpha_1$ | $\alpha_B$ |
|---|---|---|---|
| H II→H I | 4.19 | 1.59 | 2.61 |
| He II→He I | 4.37 | 1.57 | 2.81 |
| He III→He II | 21.9 | 6.54 | 15.4 |

Setting `par%recomb_model = 'hui_gnedin'` restores the earlier Hui & Gnedin (1997, MNRAS 292, 27) case-A/B fits with $\lambda = 2T_{\rm TR}/T$ ($T_{\rm TR}$ = 157807, 285335, 631515 K for H I, He I, He II):

$$\alpha_A^{\rm HII} = 1.269\times10^{-13}\,\lambda^{1.503}\big/\left[1+(\lambda/0.522)^{0.470}\right]^{1.923}, \qquad \alpha_B^{\rm HII} = 2.753\times10^{-14}\,\lambda^{1.500}\big/\left[1+(\lambda/2.740)^{0.407}\right]^{2.242},$$
$$\alpha_A^{\rm HeII} = 3.000\times10^{-14}\,\lambda^{0.654}, \qquad \alpha_B^{\rm HeII} = 1.260\times10^{-14}\,\lambda^{0.750}, \tag{8}$$

He III hydrogenic ($Z$=2). These are retained so the recorded gates reproduce; they agree with the default to $< 0.6\%$ for H II and He III, but give He II→He I 7% lower ($\alpha_B = 2.62$ vs $2.81\times10^{-13}\,{\rm cm^3\,s^{-1}}$). Recombination *cooling* uses the companion fits

$$\beta_B^{\rm HII} = 3.435\times10^{-30}\,T\,\lambda^{1.970}\big/\left[1+(\lambda/2.250)^{0.376}\right]^{3.720}\ {\rm erg\,cm^3\,s^{-1}}, \tag{9}$$

(case A analog with $1.778\times10^{-29}$, 0.541, 1.965, 0.502, 2.697), He III scaled as for $\alpha$, and He II as $k_B T\,\alpha_{A,B}^{\rm HeII}$ (their recommendation; He II dielectronic recombination is negligible at nebular temperatures).

## 3.3 H/He microphysics: comparison with other codes

The H/He cross sections and recombination can be placed next to three reference codes. For the photoionization cross section, MoCHII and EXHALE evaluate the exact hydrogenic formula [Eq. (5)] for H I/He II; MOCASSIN fits them with VFKY96 (agreeing with the exact curve to $< 1\%$), and Cloudy resolves them from its iso-electronic H-like/He-like model atoms. For the recombination that closes the ionization balance (Table 1), MoCHII's default (Badnell RR + Mao & Kaastra $\alpha_1$) and Cloudy (Badnell, level-resolved) share the modern Strathclyde data; MOCASSIN uses the older Verner & Ferland (1996) total; EXHALE, a planetary-atmosphere code, adopted the same Badnell RR + Mao & Kaastra construction as its default on 2026-07-17 (Hui & Gnedin remains a legacy option). MOCASSIN and Cloudy always work in case A and transport the diffuse ground-recombination field explicitly, whereas MoCHII defaults to case B (on-the-spot) with an optional explicit diffuse field, and EXHALE is case B only.

Table 1: H/He recombination in the ionization balance: source and $\alpha$ at $10^4$ K ($10^{-13}\,{\rm cm^3\,s^{-1}}$; case A / case B where both apply).

| code | source | H II→I | He II→I | He III→II |
|---|---|---|---|---|
| MoCHII (default) | Badnell RR + Mao & Kaastra $\alpha_1$ | 4.19 / 2.61 | 4.37 / 2.81 | 21.9 / 15.4 |
| MoCHII (`hui_gnedin`) | Hui & Gnedin 1997 | 4.30 / 2.59 | 4.23 / 2.62 | 22.3 / 15.5 |
| EXHALE (default) | Badnell RR + Mao & Kaastra $\alpha_1$ | – / 2.61 | – / 2.81 | – / 15.4 |
| EXHALE (`legacy_hhe_rates`) | Hui & Gnedin 1997 (case B) | – / 2.59 | – / 2.62 | – / 15.5 |
| MOCASSIN | Verner & Ferland 1996 (case A) | 4.19 | 4.84 | 21.9 |
| Cloudy | Badnell RR + DR (case A) | 4.19 | 4.37 | 21.9 |

The H II and He III rates agree across all four codes to within a few percent. The one substantive difference is He II→He I: MoCHII's new default (4.37) and Cloudy's Badnell value coincide, MOCASSIN's Verner & Ferland value (4.84) is the high outlier, and the earlier Hui & Gnedin (4.23) was low. Moving MoCHII onto the modern Badnell value therefore raises its He recombination by $\sim 7\%$, in the direction that reduces the He over-ionization seen relative to converged MOCASSIN in the HII40 benchmark (§5. discusses that residual).

## 3.4 Collisional ionization (Voronov 1997)

$$k(T) = A\,\frac{1+P\sqrt{U}}{X+U}\,U^K e^{-U}, \qquad U = \frac{\Delta E}{kT}, \tag{10}$$

with parameters $(\Delta E\,[\mathrm{eV}], P, A, X, K)$ parsed from the author's `cfit.f` table:

| ion | $\Delta E$ | $P$ | $A\,[\mathrm{cm^3\,s^{-1}}]$ | $X$ | $K$ |
|---|---|---|---|---|---|
| H I | 13.6 | 0 | $2.91\times10^{-8}$ | 0.232 | 0.39 |
| He I | 24.6 | 0 | $1.75\times10^{-8}$ | 0.180 | 0.35 |
| He II | 54.4 | 1 | $2.05\times10^{-9}$ | 0.265 | 0.25 |

plus the 16 registry rows in the element files. An optional `par%ci_model = 'dere_hybrid'` multiplies each Voronov rate by the constant Dere (2007)/Voronov ratio per (element, stage) taken from Cloudy c23.01's `DereRatio` table; the default `'voronov'` leaves the rate unchanged. In photoionized gas at $\sim 10^4$ K the metal collisional ionization is Boltzmann-suppressed far below photoionization, so the hybrid shifts the ionization structure only in shielded, low-ionization zones where collisions are the sole metal-ionization channel.

## 3.5 Metal recombination: Badnell RR + DR

For the registry cascade, total recombination of the recombining ion is Badnell (2006, ApJS 167, 334) radiative recombination

$$\alpha_{\mathrm{RR}}(T) = A\left[\sqrt{T/T_0}\left(1+\sqrt{T/T_0}\right)^{1-b}\left(1+\sqrt{T/T_1}\right)^{1+b}\right]^{-1}, \qquad b = B + Ce^{-T_2/T}, \tag{11}$$

plus the Badnell-group dielectronic fits $\alpha_{\mathrm{DR}}(T) = T^{-3/2}\sum_i c_i e^{-E_i/T}$. Coefficients are read from the Badnell/Strathclyde AMDPP TAMOC tables (the `clist_K` files, 2023 release, copied into `data/atomic/badnell_{rr,dr}.dat` and verified md5-identical to the copies Cloudy c23.01 redistributes), using the $M$=1 ground-level fit of each recombining ion. Most ions are numerically identical to the CHIANTI v11.0.2 `.rrparams/.drparams` (which *are* Badnell data), the notable exception being Si II: CHIANTI 11 still carries the Abdel-Naby et al. (2012) DR, whose $\sim 10^4$-K resonance is stronger, so the 2023 refit lowers the total $\mathrm{Si}^+ + e \to \mathrm{Si}^0$ recombination to $0.48\times$ the CHIANTI value at $10^4$ K. Ions Badnell does not tabulate (the low Ar/Fe/Cl/Ca stages, confirmed absent at every $M$) fall back to the CHIANTI files. Where CHIANTI carries no Badnell DR (Ar II, Ar III), the `.drparams` type-2 entries are the Shull & Van Steenberg (1982) form $\alpha_{\mathrm{DR}} = A\,T^{-3/2}e^{-T_0/T}(1+B\,e^{-T_1/T})$, written as `DR2` lines and supported by the same reader (negligible at nebular temperatures; kept for completeness). One further CHIANTI RR type appears with iron: the Fe II `.rrparams` is a type-3 power law, written as `RR2` lines and evaluated as $\alpha_{\mathrm{RR}} = A\,(T/10^4\,\mathrm{K})^{-\eta}$.

## 3.6 Charge exchange with hydrogen

First-stage reactions use the verified transcription of Huang et al. (2023, ApJ 951, 123) Table 4 (sources Stancil et al. 1998/1999, Lin et al. 2005, Zhao et al. 2005, Kingdon & Ferland 1996), e.g. for oxygen

$$k(\mathrm{O}+\mathrm{H}^+) = 2.08\times10^{-9}\,T_4^{0.405} + 1.11\times10^{-11}\,T_4^{-0.458},$$
$$k(\mathrm{O}^+ + \mathrm{H}) = \left(1.26\times10^{-9}\,T_4^{0.517} + 4.25\times10^{-10}\,T_4^{0.00669}\right) e^{-227/T}. \tag{12}$$

Higher stages ($\mathrm{X}^{i+} + \mathrm{H}^0 \to \mathrm{X}^{(i-1)+}$, recombination direction) use the Kingdon & Ferland (1996) fits $k = a\,T_4^b\,(1 + c\,e^{d\,T_4})$, transcribed from the MOCASSIN `update_mod.f90` `chex` table and evaluated at $T$ clamped into the fit validity ($\sim$5 000–50 000 K). *Stage mapping (corrected 2026-07-13).* MOCASSIN's $\mathtt{chex}(Z, \mathrm{ion})$ is the reaction $\mathrm{X}^{\mathrm{ion}+} + \mathrm{H}^0 \to \mathrm{X}^{(\mathrm{ion}-1)+}$, and its inline comment labels the *product* (lower) ion. The recombining ion of transition $i$ is $\mathrm{X}^{i+}$, so the entry is $\mathtt{chex}(Z, i)$. An earlier version read the product-labeled comment as the reactant and used $\mathtt{chex}(Z, i{+}1)$, placing every higher-stage rate one ionization stage too low; the corrected assignment (verified against MOCASSIN's own ratio loop and CMacIonize) is below.

| reaction | $a$ [$10^{-9}\mathrm{cm}^3\,\mathrm{s}^{-1}$] | $b$ | $c$ | $d$ |
|---|---:|---:|---:|---:|
| $\mathrm{C}^{2+}$ + H | $1.67\times10^{-4}$ | 2.79 | 304.72 | −4.07 |
| $\mathrm{C}^{3+}$ + H | 3.25 | 0.21 | 0.19 | −3.29 |
| $\mathrm{N}^{2+}$ + H | 0.305 | 0.60 | 2.65 | −0.93 |
| $\mathrm{O}^{2+}$ + H | 1.04 | 0.27 | 2.02 | −5.92 |
| $\mathrm{Ne}^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $\mathrm{S}^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $\mathrm{S}^{3+}$ + H | 2.29 | 0.0402 | 1.59 | −6.06 |
| $\mathrm{Ar}^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $\mathrm{Ar}^{3+}$ + H | 4.57 | 0.27 | −0.18 | −1.57 |
| $\mathrm{Si}^{2+}$ + H | 1.23 | 0.24 | 3.17 | $4.18\times10^{-3}$ |
| $\mathrm{Si}^{3+}$ + H | 0.49 | −0.0874 | −0.36 | −0.79 |
| $\mathrm{Si}^{4+}$ + H | 7.58 | 0.37 | 1.06 | −4.09 |
| $\mathrm{Mg}^{2+}$ + H | $8.58\times10^{-5}$ | $2.49\times10^{-3}$ | 0.0293 | −4.33 |
| $\mathrm{Fe}^{2+}$ + H | 1.26 | 0.0772 | −0.41 | −7.31 |
| $\mathrm{Fe}^{3+}$ + H | 3.42 | 0.51 | −2.06 | −8.99 |

Mg and Fe were already correct: their Huang Table 4 reactant labels happen to land on $\mathtt{chex}(Z, i)$. Silicon's first stage uses the Huang A9/A10 fits ($\mathrm{Si}+\mathrm{H}^+$ and $\mathrm{Si}^+ + \mathrm{H}^0$, the latter $2.371\times10^{-12}\mathrm{cm}^3\,\mathrm{s}^{-1}$ at $10^4$ K); Cl and Ca have no KF96/Huang transcription. The Mg and Fe first-stage reactions follow Huang Table 4: Mg + $\mathrm{H}^+$ is the KF96 form with an added reverse barrier $e^{-6.91/T_4}$ (endothermic); Fe + $\mathrm{H}^+$ is a fast constant $4.0\times10^{-9}\mathrm{cm}^3\,\mathrm{s}^{-1}$ with reverse $a\,(T/300)^b\,e^{-6.61/T_4}$. The correction raises the HII40 doubly/triply-ionized fractions (Section 5.4); the earlier good agreement of some metal fractions with the underconverged MOCASSIN baseline was partly fortuitous.

## 3.7 Cooling fits

Collisional line cooling for the thermal loop uses the EXHALE multi-exponential form fitted to direct CHIANTI v11.0.2 evaluations (`fit_cooling.py`; reader `chianti_cooling.py` parses `.elvlc`/`.scups` with Burgess & Tully 1992 descaling):

$$\Lambda_{\rm ion}(T) = T^{-1/2}\sum_{i=1}^{N} A_i\, e^{-T_i/T} \qquad [\mathrm{erg\,cm^3\,s^{-1}\ per\ }(n_e n_{\rm ion})], \tag{13}$$

evaluated in the optically thin $n_e \to 0$ limit of statistical equilibrium: every ion sits in its lowest level, excitation happens out of that level only, and the cascade radiates the full excitation energy, so $\Lambda_{\rm ion}$ is the excitation power out of the ground level and equals what the $n$-level solve (Section 3.8) returns as $n_e \to 0$. Transition energies are the observed `.elvlc` level differences in both the Boltzmann factor and the radiated photon (the `.scups de` is the Burgess–Tully scaling energy, and detailed balance ties the excitation rate to the level gap that appears in the equilibrium matrix). Fit range $10^3$–$10^5$ K; Levenberg–Marquardt on $\log\Lambda$ with the $T_i$ ladder derived from the excitation temperatures $\Delta E_{1u}/k$ of each ion.

| ion | $N$ | max err | ion | $N$ | max err |
|---|---|---|---|---|---|
| H I | 4 | 0.52% | O I | 6 | 0.20% |
| C I | 6 | 0.27% | O II | 4 | 0.27% |
| C II | 5 | 0.35% | O III | 5 | 0.32% |
| C III | 6 | 0.50%[a] | S II | 3 | 2.23% |
| N I | 5 | 0.58% | S III | 5 | 0.23% |
| N II | 5 | 0.66% | S IV | 3 | 0.61% |
| N III | 5 | 0.70% | Ne II | 5 | 0.40% |
| | | | Ne III | 5 | 0.34% |
| | | | Ar III | 2 | 0.22%[b] |
| | | | Ar IV | 4 | 0.27% |

[a]in the relevant range ($\Lambda > 10^{-3}$ max); 5.5% at the negligible low-$T$ tail. [b]at $10^4$ K; 3.9% at the $10^5$-K endpoint.

By construction Eq. (13) carries no collisional de-excitation, so it is exact for lines whose critical density is far above the local $n_e$ and increasingly too large for those below it — the low-$n_{\rm crit}$ lines [C II] 158 $\mu$m, [O III] 52/88 $\mu$m and the [O II]/[S II] optical doublets, whose electron critical densities are $10^2$–$10^{3.5}\,\mathrm{cm^{-3}}$, so at the $n_e \sim 10^2\,\mathrm{cm^{-3}}$ of an H II region they overcool the line budget by $\sim$12% (measured against a CLOUDY c25.00 cooling decomposition on a fixed state). The thermal loop therefore treats Eq. (13) as the $n_e \to 0$ *carrier* and suppresses it to the local density. A $(T, n_e)$ table, built once at setup (`nlevel_cooling_mod`) by solving the same $n$-level atom over a grid ($\log_{10} T = 2$–6, $\log_{10} n_e = -6$–10), stores for each ion the suppression ratio

$$S_{\rm sup}(T,n_e) = \frac{\Lambda_{\rm SE}(T,n_e)}{\Lambda_{\rm SE}(T,n_e\to 0)} \in (0,1], \qquad \Lambda_{\rm SE}(T,n_e) = \frac{1}{n_e}\sum_k n_u^{(k)} A_k\,\Delta E_k, \tag{14}$$

and the low-density reference $\Lambda_{\rm SE}(T, n_e \to 0)$; the runtime cooling coefficient combines the two as

$$\Lambda(T,n_e) = \Lambda_{\rm fit}(T)\big[f_{\rm cov}(T)\,S_{\rm sup}(T,n_e) + (1 - f_{\rm cov}(T))\big], \qquad f_{\rm cov} = \min\Big(1,\ \tfrac{\Lambda_{\rm SE}(T,n_e\to 0)}{\Lambda_{\rm fit}(T)}\Big). \tag{15}$$

Here $f_{\rm cov}$ is the fraction of the low-density cooling the (often truncated) $n$-level model resolves; the remainder $1 - f_{\rm cov}$ — the high-excitation resonance lines, of far higher critical density — is left in its low-density limit, so a truncated model cannot bias the total. As $n_e \to 0$, $S_{\rm sup} \to 1$ and $\Lambda \to \Lambda_{\rm fit}$ *exactly*, and an ion with no $n$-level model would keep the plain fit — there is none left, since C I and N I, the last two gaps, now carry models of their own (their far-infrared fine-structure lines have $n_{\rm crit} \sim 2$–$10\,{\rm cm}^{-3}$, so the suppression is severe: $S_{\rm sup} = 0.43$ at $10^4$ K and $n_e = 10^4\,{\rm cm}^{-3}$). This density-suppressed cooling (`par%cooling_model = local_ne`, the default) is built from the very solve that produces the diagnostic lines (Section 3.8), so the thermal balance and the emergent line list are one physics; the pure zero-density fit is retained as the `low_density` option, exact in the diffuse gas the limit is written for. The density-dependent emissivities themselves come from the $n$-level data.

## 3.8 $n$-level (diagnostic) data

For output-time diagnostics, `fit_nlevel.py` exports for each ion the lowest $n$ levels (energies, statistical weights), the radiative $A$-values (`.wgfa`), and the effective collision strengths $\Upsilon(T)$ as low-order Chebyshev fits in Burgess–Tully scaled space: with $e_t = kT/\Delta E$,

$$s = \begin{cases} 1 - \ln C/\ln(e_t + C) & \text{types 1, 4} \\ e_t/(e_t + C) & \text{types 2, 3, 5, 6} \end{cases}, \qquad u = \frac{2(s - s_{\rm lo})}{s_{\rm hi} - s_{\rm lo}} - 1, \qquad y = \sum_k c_k T_k(u), \tag{16}$$

descaled by type ($y\ln(e_t{+}e)$; $y$; $y/(e_t{+}1)$; $y\ln(e_t{+}C)$; $y/e_t$; $10^y$), with a log flag storing $\log_{10} y$ for weak peaked intercombination transitions. The $n$-level export now covers the 24 line-emitting ions of the eleven-element registry (Table 2); the worst transition fit error is 1.7% (O III $^1S_0$–$^5S_2$, $\Upsilon \sim 10^{-4}$), typically $\lesssim 0.5\%$. Ar II has no level/collision data in CHIANTI v11 (only RR/DR), so that stage is tracked without cooling or lines ([Ar II] 6.98 $\mu$m omitted). Verification is layered: the fitted files reproduce direct CHIANTI spline evaluation of the classic ratios to 0.05–0.5%; the PyNeb cross-check gives [O III] 5007/4363 within 1.2% and [N II] 6584/5755 within 0.3% (the density diagnostics differ by 6–10% at $10^4\,{\rm cm}^{-3}$ from PyNeb's independent collision data — database spread); and the Fortran solver (Section 4.14) matches the Python reference implementation to four significant digits on 26 lines.

Table 2 lists the principal collisional diagnostic lines each metal block produces (the complete transition list for each ion is written to `<base>_lines.txt`); only active-registry ions with an `nlevel_<el>_<stage>` file get a block. The H I/He II/He I recombination-line set is documented separately in the recombination-lines subsection below.

Table 2: Metal collisional diagnostic lines from the $n$-level solve (block `emis_<el>_<stage>` in `<base>_emis`; principal lines, Å unless $\mu$m noted). C, N, O, Ne, S are active by default; Ar, Mg, Fe, Si, Cl, Ca are activated by `par%abund_<El>`.

| ion (block) | principal lines |
|---|---|
| c_2 [C II] | 2325 (UV multiplet), 158 $\mu$m |
| c_3 C III] | 977, 1907, 1909, 178 $\mu$m |
| n_2 [N II] | 3064/3071, 5756, 6550, 6585, 122 $\mu$m, 205 $\mu$m |
| n_3 [N III] | 1750 (UV multiplet), 57 $\mu$m |
| o_1 [O I] | 5579, 6302, 6366, 63 $\mu$m, 145 $\mu$m |
| o_2 [O II] | 2471, 3727, 3730, 7321/7331 |

*Table 2 (continued)*

| ion (block) | principal lines |
|---|---|
| o_3 [O III] | 1661/1666, 2322/2332, 4364, 4960, 5008, 52 $\mu$m, 88 $\mu$m |
| ne_2 [Ne II] | 12.8 $\mu$m |
| ne_3 [Ne III] | 3343, 3870, 3969, 15.55 $\mu$m, 36 $\mu$m |
| s_2 [S II] | 4070/4078, 6718, 6733, 1.03 $\mu$m, 315 $\mu$m |
| s_3 [S III] | 3722, 6314, 8831, 9070, 9532, 12 $\mu$m, 18.7 $\mu$m, 33.5 $\mu$m |
| *Optional — activate with* par%abund_<El> | |
| ar_3 [Ar III] | 3006/3110, 5193, 7138, 7753, 8.99 $\mu$m, 21.8 $\mu$m |
| ar_4 [Ar IV] | 2854/2869, 4713, 4741, 7170 (multiplet), 56 $\mu$m, 78 $\mu$m |
| mg_2 Mg II | 2796, 2804 |
| fe_2 [Fe II] | 1.26/1.32 $\mu$m, 5.34 $\mu$m, 17.9 $\mu$m, 26 $\mu$m (40 lines total) |
| fe_3 [Fe III] | 4659, 4703, 4881, 5011, 22.9 $\mu$m (38 lines; 16/34-level with fe_levels_full) |
| si_2 [Si II] | 2335 (multiplet), 34.8 $\mu$m |
| si_3 Si III] | 1206, 1883, 1892, 38 $\mu$m |
| si_4 Si IV | 1394, 1403 |
| cl_2 [Cl II] | 3587/3679, 6164, 8581, 9126, 14.4 $\mu$m, 33 $\mu$m |
| cl_3 [Cl III] | 3344/3354, 5519, 5539, 8480 (multiplet) |
| cl_4 [Cl IV] | 3120/3205, 5325, 7263, 7533, 11.8 $\mu$m, 20.3 $\mu$m |
| ca_2 Ca II | 3935, 3970, 7293/7326, 8500/8544/8665 |
| ca_5 [Ca V] | 3999, 5311, 6088, 3.05 $\mu$m, 4.16 $\mu$m, 11.5 $\mu$m |

Some registry ions are tracked in the ionization balance but carry no line block because CHIANTI v11 has no level/collision data for them: Ar II, Cl V, Ca III, Ca IV.

## 3.9 Free-free Gaunt factors and continuum data

`gaunt.f90` is the verbatim MOCASSIN copy of the Hummer (1988) two-dimensional Chebyshev fit of the Karzas & Latter (1961) free-free Gaunt factor ($\leq 0.7\%$; interface `getGauntFF(z, log10Te, xlf, g)`). The thermal loop uses the thermally averaged $\bar{g}(T,Z) = \int g_{\rm ff}(u)\, e^{-u}\, du$ tabulated at setup ($\bar{g}(10^4\,{\rm K}, Z{=}1) = 1.2626$), now over net charge $Z = 1$–$4$ so the trace-metal free-free (`par%metal_freefree`) has its Gaunt factor. The van Hoof et al. (2014) table is now available as a drop-in (`par%gaunt_vh14`; `data/gauntff_vh14.dat`, an $81 \times 146$ grid in $\log_{10}\gamma^2 = \log_{10}(Z^2\,{\rm Ry}/kT)$ and $\log_{10} u = \log_{10}(h\nu/kT)$, selected at both call sites — the cooling average and the continuum $\gamma_{\rm ff}$). The two compilations agree: switching moves $T_e$ by a median $5\times10^{-5}$, at most 0.5%, consistent with the sub-percent Hummer accuracy van Hoof et al. report. Hummer remains the default so the recorded gates reproduce.

The nebular continuum tables `data/gamma{HI,HeI,HeII}.dat` (MOCASSIN inheritance; 21 temperatures $\times$ paired-edge frequency points) were *numerically verified* (2026-07-12) against a hydrogenic Kramers Milne sum: they contain free-bound *plus* free-free combined coefficients (at $\nu = 0.0069$ Ryd, $10^4$ K the table value 9.47 decomposes into ff $\simeq 8.9$ + fb $\simeq 0.6$ in $10^{-40}$ units; near the Balmer edge the residual is $\sim$10%, the size of the neglected bound-free Gaunt factor).

Consequently the port never adds free-free inside the table range. The Ercolano & Storey (2006) item is resolved as *provenance*: the MOCASSIN 2.02 tables ship in the ES06 compact paired-edge format, so no swap is needed — the verification above is the record. Two-photon continua follow MOCASSIN: Nussbaumer & Schmutz (1984) spectral shape with Pengelly (1964) $\alpha_{\rm eff}(2s)$ and the Osterbrock $q(2^2{\rm S}{-}2^2{\rm P})$ collisional suppression for H I; the $Z$-scaled form with Storey & Hummer $\alpha_{\rm eff}$ for He II; Almog & Netzer (1989) shape (`data/HeI2phot.dat`) with Benjamin, Skillman & Smits (1999) $\alpha_{\rm eff}$ for He I.

## 3.10 H I and He II recombination lines (Storey & Hummer 1995)

`make_sh95_lines.py` parses the SH95 case-B data (`e1bx.d`) with the reference reader indexing $k = (2n_{\rm top} - n_{ll} - n_l + 1)(n_l - n_{ll})/2 + n_{\rm top} - n_u + 1$ and exports H$\alpha$, H$\beta$, H$\gamma$, H$\delta$, P$\alpha$, P$\beta$, Br$\gamma$ on the full $(T, n_e)$ grid (11 × 13) to `data/atomic/sh95_hi_caseB.txt`; Fortran interpolates bilinearly in $(\log T, \log n_e)$. Checks: $4\pi j_{\rm H\beta}/(n_e n_p) = 1.235\times10^{-25}$ at $(10^4\,{\rm K}, 10^2\,{\rm cm}^{-3})$ and H$\alpha$/H$\beta$ = 2.864 (canonical values).

He II ($Z = 2$) comes from the *original* CDS file `e2b.d`, whose format differs from the repackaged `e1bx.d` (six-token block headers; index $k = (n_{\rm cut} - n_u)(n_{\rm cut} + n_u - 1)/2 + n_l$ with $n_{\rm cut} = 25$, from the SH95 `intrat.f` reader): six lines (1640, 4687, 3204, 10126, 5413, 4543 Å) on a 12 × 13 grid in `data/atomic/sh95_heii_caseB.txt`, weighted in the line table by $n_e n_{\rm HeIII}$. Checks: $4\pi j_{4687}/(n_e n_{\rm HeIII}) = 1.495\times10^{-24}$ at $(10^4\,{\rm K}, 10^4\,{\rm cm}^{-3}) = \alpha^{\rm eff}_{4687}(3.57\times10^{-13})\,h\nu$, and $1640/4687 = 6.56$ there.

He I lines use the same interpolator (a third table in `sh95_mod`) but a different source: the Porter et al. (2012, 2013) case-B emissivities, parsed by `make_hei_lines.py` from Cloudy c23.01's `he1_case_b.dat` (copied into `data/atomic`) onto a 21 × 14 $(T, n_e)$ grid. Unlike the H I and He II tables, these are case-B collisional-radiative TOTALS, not pure recombination cascades: the tabulated $4\pi j/(n_e n_{\rm He^+})$ is density-dependent (the 10833 Å coefficient rises 6.6x from $n_e = 10$ to $10^4$ cm$^{-3}$ at $10^4$ K) because it already includes collisional excitation out of the $2^3$S metastable. Eight H II-region diagnostics are exported (3890, 4027, 4473, 5877, 6680, 7067, 7283, 10833 Å, vacuum), weighted in the line table by $n_e n_{\rm HeII}$ (He$^+$ recombining). Emergent ratios are physical: $5877/4473 = 2.88$, $6680/4473 = 0.82$, $7067/4473 = 0.49$ at $\sim$7 kK.

The H I and He II tables follow `par%case_ab`: case-A tables (`sh95_{hi,heii}_caseA.txt`, parsed from the original CDS files `e1a.d`/`e2a.d` onto the 10 × 9/12 × 9 grids case A tabulates) are loaded when the ionization runs case A, so the line emissivities match the balance; case B (H$\alpha$/H$\beta$ = 2.94) is the default. The He I Porter set is case B only. An optional collisional H I Balmer component (`par%h_coll_effects`, off by default) solves a 25-level H I atom (from CHIANTI `h_1` through the $n$-level pipeline) with collisional excitation from the ground level, groups the $l$-resolved emission into H$\alpha$/H$\beta$/H$\gamma$, weights it by $n_{\rm HI}$, and writes it as separate `hc` rows — a first-cut estimate (optically-thin cascade) for the front and hot-cell bias, kept apart from the recombination lines.

## 3.11 He I $2^3$S metastable diagnostic

The Porter case-B 10833 Å table above is a collisional-radiative *total* emissivity, correct in the dense, optically-thin, recombination-dominated H II regime it was built for. It does *not* give the $2^3$S population $n_3$, its line-of-sight column, or the 10833 Å *absorption* opacity — the observables of exoplanet transit, chromospheric, and PDR 10833 work, where the line is seen against a background. `hei_metastable_mod` (`par%hei_metastable`) adds these as a separate diagnostic; it does not add or modify any emission row, and in particular *nothing is summed onto* the Porter 10833 line (which already carries the metastable-collisional term — summing an explicit collisional 10833 would double-count).

**The $2^3$S balance (Oklopcic & Hirata 2018).** OH18 track the neutral singlet ground $1^1$S, the metastable triplet $2^3$S, and the ion. Because $n_3/n_{\rm He} \sim 10^{-7}$ the balance is linear in $n_3$ — a closed-form quotient in each leaf, no iteration:

$$n_3 = \frac{\alpha_3\, n_e\, n_{\rm HeII} + q_{13}\, n_e\, n_{\rm HeI}}{A_{31} + \Phi_3 + n_e\,(q_{31a} + q_{31b}) + n_{\rm HI}\, Q_{31}}, \tag{17}$$

with source terms recombination into the triplet system ($\alpha_3\, n_e\, n_{\rm HeII}$) and e-impact excitation $1^1{\rm S} \to 2^3{\rm S}$ ($q_{13}\, n_e\, n_{\rm HeI}$), and sinks the radiative decay $A_{31}$ ($2^3{\rm S} \to 1^1{\rm S}$), photoionization $\Phi_3$, spin-changing e-impact transfer into the singlets ($q_{31a} = 2^3{\rm S} \to 2^1{\rm S}$, $q_{31b} = 2^3{\rm S} \to 2^1{\rm P}$ — *not* $2^3{\rm S} \to 1^1{\rm S}$), and Penning/associative ionization with neutral hydrogen ($n_{\rm HI}\, Q_{31}$). Two limits anchor the gates: the low-density radiative limit $n_3/n_{\rm He^+} = \alpha_3 n_e/A_{31}$ and the high-density collisional plateau $\alpha_3/(q_{31a}+q_{31b})$, crossing at $n_{\rm crit} = A_{31}/(q_{31a}+q_{31b})$.

The rate coefficients are closed-form fits transcribed from the EXHALE triplet implementation (offline into the coefficient file, Fortran only evaluates): $\alpha_3(T) = 2.10\times10^{-13}(T/10^4)^{-0.778}$ (OH18); $q_{13}, q_{31a}, q_{31b}$ from Bray et al. (2000) collision strengths via OH18; $A_{31} = 1.272\times10^{-4}$ s$^{-1}$ (the CHIANTI `he_1.wgfa` $A(1\text{–}2) = 1.73\times10^{-4}$ is $\sim$36% high and is *not* used); $Q_{31}$ from Taylor et al. (2025) Table 2 (Garcia Muñoz 2025 recorded as the alternative, 1.2–2.2$\times$ lower — the documented uncertainty of the neutral/PDR sink). A consistency check: at $10^4$ K, $\alpha_3 = 0.998 \times 0.75\, \alpha_B$(He II), matching the `HEI_F3S` = 0.75 triplet branching the diffuse field already uses.

**The $2^3$S photoionization rate.** $\Phi_3 = \sum_\nu 4\pi J_\nu\, \sigma_3(\nu)\, \Delta\nu/(h\nu)$ over the *full* band (the reduced tally `jt_ion`, the same integral pattern as $\Gamma_i$). The threshold is 4.78 eV, but $\sigma_3 \sim 1\text{–}2\times10^{-18}$ cm$^2$ remains significant far above 13.6 eV (Cooper minimum near 34.7 eV, bump near 45.6 eV), so the ionizing band alone gives $\Phi_3 > 0$ wherever a field exists; `add_fuv` with `efuv_min`= 4.78 adds the soft-UV part $[4.78, 13.6)$ eV (a rank-0 note fires when the band floor sits above 4.78 eV). $\sigma_3(E)$ is two VFKY96 wings joined by a log-linear bridge, a $\sim$4% fit to Norcross (1971) + TOPbase.

**10833 Å absorption (Stage 2).** From $n_3$ the line-center opacity of each fine-structure component is

$$\kappa_{0,i} = \frac{\sqrt{\pi}\, e^2}{m_e c}\, f_i\, \frac{\lambda_i}{b}\, n_3, \qquad b = \sqrt{2kT_e/m_{\rm He} + v_{\rm turb}^2}, \tag{18}$$

the classical integrated cross section $\pi e^2/(m_e c) = 0.02654$ cm$^2$ Hz divided by $\sqrt{\pi}\, b$ for the Gaussian peak. The three $2^3{\rm S} \to 2^3{\rm P}_J$ components (vacuum 10832.06 / 10833.22 / 10833.31 Å; $f$ = 0.0599/0.1797/0.2996, ratio 1:3:5; $A$ = $1.0216\times10^7$ s$^{-1}$ each) take the NIST ASD values (Drake 2007), cross-checked against p-winds; the CHIANTI `he_1.wgfa` $A \simeq 1.15\times10^7$ ($\sim$13% high) is *not* used. The $J' = 1, 2$ pair sits 0.089 Å apart, far inside the $\sim$0.4 Å thermal Doppler width at $10^4$ K, so they blend into the observed "doublet" (their line-center opacities are summed); $J' = 0$ at 10832.06 Å is resolved. The Python `HeiMetaData` reader deposits $n_3$ and $\kappa_0$ along the line of sight (the flux-conserving deposit of the emissivity maps) into column [cm$^{-2}$] and $\tau_0 = \int \kappa_0\, {\rm d}l$ maps, and warns when line-center $\tau_0 > 1$.

The module runs once at output time (no hot-loop cost, no feedback — $n_3$ is a $\sim10^{-7}$ trace of He). Caveats recorded at the code site: (i) *local equilibrium* — the advection term $v\, {\rm d}f_3/{\rm d}r$ is absent, valid where the $2^3$S destruction time is short against any flow time (dense H II), not a faithful outflow benchmark; (ii) 10833 can be optically thick in dense metastable columns, where the optically-thin $\tau_0$ diagnostic breaks down (line transfer is out of scope).

# 4. Algorithms

## 4.1 The iteration loop

read grid → persist gas leaf state → fill `kap_ion`
**do** iter = 1, `gas_niter`
  zero the band tally $J_b$
  transport stellar packets (analytic edge walk, Eq. 3)
  [diffuse] rebuild the diffuse emission table from the current state;
    emit and transport $L_{\rm dif}/L_{\rm pkt}$ diffuse packets
  ALLREDUCE $J_b$; rate integrals (Eq. 4); metal Γ's
  solve each leaf: ionization only or coupled $(x, T_e)$
  refill `kap_ion` from the new state (the stellar tally is
    recomputed from zero each iteration — no reuse)
  converged by `par%conv_crit`: cell-max or volume-integrated tests (below)
**end do**
outputs: rates + state maps; line luminosities; nebular continuum

Every rank runs the identical cell solves (inputs are the reduced tallies), so all ranks agree on convergence; only the node master writes the shared state arrays. Under-relaxation on $x$ (`par%ion_relax`) is available for sharp I-fronts. Practical lessons recorded: a fully neutral start advances the front only ∼one cell per iteration (the photons die in the first neutral cell), while an ionized start converges globally; and max $|\Delta x_{\rm HII}|$ stalls at the front-cell Monte-Carlo noise floor (∼2–4×$10^{-2}$ at 4×$10^7$ packets), so tolerances below that are unreachable by that criterion — the diffuse field, interestingly, smooths the front and converges to 3×$10^{-3}$.

**Convergence measures.** Two pairs are computed every iteration and `par%conv_crit` selects which gates the loop: the cell-max tests (max $|\Delta x_{\rm HII}| <$ `gas_tol`, max $|\Delta T_e|/T_e <$ `gas_tol_te`) and the volume-integrated tests

$$\frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HII}|}{\sum_{\rm leaf} V\,x_{\rm HII}} < \texttt{gas\_tol}, \qquad \frac{\sum_{\rm leaf} V\,n_e\,|\Delta T_e|}{\sum_{\rm leaf} V\,n_e\,T_e} < \texttt{gas\_tol\_te}. \tag{19}$$

The volume measures fix both stall modes of the cell-max tests: front jitter occupies little volume (the front shell is a few thousand cells against ∼$10^5$ ionized ones, so its noise enters the $L_1$ norm suppressed by that ratio), and the $n_e$ weight removes the vacuum and no-heating cells whose `te_min`-pinned oscillation dominates max $|\Delta T_e|$ in FUV/PDR configurations. `'vol'` is the recommended production setting; `'cell'` (the historical behavior) remains the default so the recorded gates reproduce. Measured on the FUV dusty Strömgren case at 8×$10^6$ packets (where the cell-max tests sat at 0.1 and 6.4 indefinitely): the volume measures decay smoothly to well-defined Monte Carlo floors of 2.7×$10^{-3}$ ($x$) and 6.6×$10^{-3}$ ($T_e$) by iteration ∼27, and running 30 further iterations leaves the solution unchanged ($V_{\rm ion}$ to 0.014%) — the floor IS convergence. Set the tolerances just above the floor for the packet count in use (here 5×$10^{-3}$ and $10^{-2}$

would stop the loop at iteration $\sim$23); the floors scale as $1/\sqrt{N_{\rm packets}}$. With `par%require_convergence` (default off) a non-converged gas iteration prints a warning and, when the switch is set, aborts before writing output; either way the rates-file header records the convergence status in the keywords `CONVERGD`, `NITERDON`, `FINALDX`, and `FINALDTE`.

## 4.2 Quasi-random photon launching (optional)

With `par%launch_sequence = 'sobol'` the packet launch draws its variables — source component, frequency bin, direction, and any entry-surface coordinates — from an Owen-scrambled Sobol net (`src/qmc_mod.f90`) instead of the Mersenne Twister, while every post-launch transport event stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count, and `par%qmc_seed` is held fixed across the nonlinear iterations (common random numbers). Because the band has no scattering, the direct field is an analytic function of the launch variables (Eq. 3), so a low-discrepancy launch set reduces its noise directly: on the fixed-state analytic-attenuation gate the cell-wise $\Gamma_{\rm HI}$ error scales as $\sim 1/N$ (Monte Carlo $1/\sqrt{N}$), with effective photon gains of 29$\times$ at $N = 2^{17}$ and 222$\times$ at $2^{20}$, at unchanged wall time and no significant bias. The method, the fixed dimension layouts, and all gates are documented in the companion memo `MoCHII_QMC`.

## 4.3 Ionization equilibrium

Per leaf at temperature $T$ (`solve_ion_cell`): with $R_i = \Gamma_i + n_e C_i(T)$ the ionization rates and $\alpha_i$ the case-A/B recombination rates, $x_{\rm HI} = \alpha n_e/(R_{\rm HI} + \alpha n_e)$ and the He stage chain $n_{i+1}/n_i = R_i/(\alpha_{i+1} n_e)$, closed by $n_e = n_{\rm H}[x_{\rm HII} + y_{\rm He}(x_{\rm HeII} + 2x_{\rm HeIII})]$ and solved by a damped fixed point on $n_e$ (mixing $\frac{1}{2}$, converged at $10^{-10}$, $\leq 200$ iterations).

## 4.4 Thermal balance

When `par%solve_te` is set, `gas_thermal_update` replaces the fixed-$T$ `gas_equilibrium_update` (§4.3) and updates each leaf's temperature once per iteration, from that iteration's radiation field, by finding the equilibrium $\mathcal{H}(T) = \mathcal{C}(T)$ — the root of the net rate

$$\mathcal{N}(T) = \mathcal{H}(T) - \mathcal{C}(T), \qquad \mathcal{H} = n_{\rm HI}\mathcal{H}_{\rm HI} + y_{\rm He} n_{\rm H}(x_{\rm HeI}\mathcal{H}_{\rm HeI} + x_{\rm HeII}\mathcal{H}_{\rm HeII}), \tag{20}$$

with the photoheating rates $\mathcal{H}_i$ (per atom) from Eq. (4). The ionization state is *re-solved* (`solve_ion_cell`, §4.3) at every trial $T$: both $\mathcal{H}$ and $\mathcal{C}$ depend on the fractions $(x_{\rm HI}, x_{\rm HeI}, x_{\rm HeII}, n_e)$, which themselves depend on $T$. The ionization response is mild, so $\mathcal{N}(T)$ is monotone through its root and bisection on $\log T_e$ in `[te_min, te_max]` is safe: each step takes the geometric midpoint $\sqrt{T_{\rm lo} T_{\rm hi}}$ (up to 60 steps, stopping at $T_{\rm hi}/T_{\rm lo} - 1 < 10^{-5}$), and if cooling already wins at `te_min` ($\mathcal{N} \leq 0$) the leaf is pinned to `te_min` while if heating still wins at `te_max` it is pinned to `te_max`.

The cooling $\mathcal{C}(T)$ (`cooling_mod`) assembles recombination (Hui–Gnedin $\beta$), free-free ($1.42554\times10^{-27}\sqrt{T}\,Z^2\,\bar{g}\,n_e n_{\rm ion}$, summed over H II, He II ($Z = 1$) and He III ($Z = 2$) with $Z$ the *net* ion charge; under `par%metal_freefree` (on by default) the trace metals are added with $Z_{\rm ion}$ = (stage −1), not the nuclear $Z$, using $\bar{g}(T, Z_{\rm ion})$ tabulated for $Z = 1$–4 — singly-ionized metals dominate this small $\sim10^{-4}$-of-the-H/He term), collisional ionization ($k_{\rm ci}\times$ ionization potential), H I line cooling, and the trace-metal sum $\sum n_e\, n_{\rm ion} \Lambda_{\rm ion}(T, n_e)$ with the cascade fractions (§4.5) re-evaluated at each trial $T$. The line-cooling coefficient $\Lambda_{\rm ion}(T, n_e)$ is the density-suppressed value of Eq. (15) by default (`par%cooling_model`

= `local_ne`), the plain fit $\Lambda_{\rm ion}(T)$ under `low_density`. PDR runs add three optional terms to $\mathcal{N}$: metal photoheating (`par%metal_heat`); grain photoelectric heating with grain recombination cooling (Bakes & Tielens 1994, `par%grain_pe`, driven by the local FUV field $G_0$, both weighted by $x_{\rm HI}$ and evaluated at $\min(T, 10^4\,{\rm K})$ so the fits stay in range — two guards the PDR gate needed against a 12–35 kK runaway); and the H-impact [C II] 158 $\mu$m / [O I] 63 $\mu$m fine-structure cooling that balances the photoelectric heating in the mostly neutral gas.

Under `par%use_sec_ion` (Shull & van Steenberg 1985) a photoelectron of kinetic energy $E_0 = h\nu - E_{{\rm th},s}$ from absorber $s$ (H I, He I, He II) with $E_0 > 40$ eV deposits only $f_{\rm heat}(x)$ of its excess as heat (below the threshold it thermalizes fully), so the photoheating integrand is weighted by $f_{\rm heat}(x)$; the un-thermalized balance drives $f_{\rm ion,HI}(x)E_0/E_{\rm th,HI}$ secondary H I and $f_{\rm ion,HeI}(x)E_0/E_{\rm th,HeI}$ secondary He I ionizations that add to the corresponding photoionization rates in `solve_ion_cell`. Here $x = (n_{\rm HII} + n_{\rm HeII} + n_{\rm HeIII})/(n_{\rm H} + n_{\rm He})$ is the ionized fraction of the H+He nuclei and the partitions are $f_{\rm heat} = 0.9971[1-(1-x^{0.2663})^{1.3163}]$, $f_{\rm ion,HI} = 0.3908(1-x^{0.4092})^{1.7592}$, $f_{\rm ion,HeI} = 0.0554(1-x^{0.4614})^{1.6660}$. The effect vanishes in fully ionized gas ($f_{\rm ion} \to 0$ as $x \to 1$, so the HII-region gates are unchanged) and matters only in partially neutral gas under a hard field (fronts, PDR, X-ray/AGN); metals are neglected as absorbers (trace). Default is off.

The solved $T_e$ is written directly (only the ionization fractions carry the under-relaxation `par%ion_relax`), $n_e$ is rebuilt from the new state — including the trace-metal electrons under `par%metal_ne` — and, as in the ionization-only path, only the node-local rank 0 runs the bisection (the reduced band tally is identical on every node master) and writes the shared state, with the convergence metrics broadcast within the node. The loop is gated by the $T_e$ convergence measures defined above (`gas_tol_te`, cell-max or volume-integrated per `par%conv_crit`).

## 4.5 The trace-metal cascade

The registry (`species_mod`) activates an element when its abundance is positive. In the trace approximation (no feedback on $n_e$ or opacity), stage fractions follow the product chain

$$\frac{n_{i+1}}{n_i} = \frac{\Gamma_i + n_e C_i + n_{\rm HII}\, k^{\rm CX}_{{\rm ion},i}}{n_e\,[\alpha_{\rm RR} + \alpha_{\rm DR}]_{i+1} + n_{\rm HI}\, k^{\rm CX}_{{\rm rec},i+1}}, \tag{21}$$

with metal $\Gamma_i$ computed once per iteration from the same band tally. Adding an ion is one entry in the data generators plus its cooling and $n$-level files.

## 4.6 Explicit diffuse field

Case B on the spot is replaced by case A rates plus explicit emission of ground-recombination packets: three channels with rates $\alpha_1 = \alpha_A - \alpha_B$ and photon energies $E = E_{\rm th} + kT_e x$, $x \sim {\rm Exp}(1)$ (the near-threshold Milne shape at bin resolution). Each iteration the emission table $L_{\rm ch}({\rm leaf}) = n_e n_{\rm rec} \alpha_1(T)(E_{\rm th} + kT)V$ is rebuilt from the current state; packets carry the stellar $L_{\rm pkt}$, sample the leaf from a CDF, a uniform position within it, and are transported by the same edge walk. At HII40 the diffuse band luminosity converges to $6.8\times10^{38}$ erg/s $= 0.50\,L_\star$. The sampled photon energy is mapped to its band bin by a binary search of the stored bin-edge array (`ion_eedge`, `ion_bin_of`), correct on both the aligned (default) and the legacy log grids: an energy exactly on a threshold edge falls in the bin *above* it, the physically correct side for a recombination photon emitted at threshold$+\delta$. The former closed-form log mapping assumed the log grid and mis-binned threshold photons once `ion_align_edges` became the default — a He I ground-recombination photon

(24.6 eV) landed in the bin *below* the He I edge, where it could not ionize helium; the fix lowers the mean $x_{\rm HeI}$ by 50.5% on a diffuse Strömgren sphere.

With `par%hei_diffuse` a fourth channel adds the He I *excited-level* recombination radiation (previously those photons simply vanished): all case-B He II recombinations cascade to $n = 2$ and decay with the AGN3 low-density branching — $3/4$ to $2^3$S (a single 19.82-eV photon), $1/4 \times 2/3$ to $2^1$P (584 Å = 21.22 eV), $1/4 \times 1/3$ to $2^1$S (two-photon; probability 0.56 of an H-ionizing photon, sampled flat in $[13.6, 20.62]$ eV). At HII40 this raises the diffuse luminosity by 4.7% and $T_e$ by 1.1% while leaving the He I/II split unchanged to four decimals — correct energy bookkeeping, and the demonstration that the He-structure residual against MOCASSIN does NOT come from this channel (the split is set by the $>$24.6-eV budget).

## 4.7 Solution-driven I-front re-refinement

At iteration `refine_iter` the octree is rebuilt from the current solution: an old leaf is flagged as front when $\epsilon < x_{\rm HI} < 1-\epsilon$ or a face neighbor differs by more than `refine_dx`; the new tree carries level `refine_lmax` wherever a cell overlaps a front-flagged leaf and `refine_lbase` elsewhere. The gas state and `rhokap` map over by position (each new leaf samples the old leaf containing its center), and every dependent array is recreated (the band opacity, tallies, metal $\Gamma$ blocks, diffuse tables, rate arrays — the last was the one initially missed, an allocate-once array that faulted when $n_{\rm leaf}$ grew). All ranks rebuild identically from the shared state, and the old grid's shared-memory windows are *recycled*: `destroy_shared_mem` (the collective counterpart of `create_shared_mem`) frees the 18 old windows right after the position sampling — the old tree's last reader — so repeated re-refinement neither leaks nor holds old and new grids in memory simultaneously.

## 4.8 Dust in the ionizing band

The three parts of the dust coupling, now complete:

- **Absorption competition.** With `par%ion_add_dust`, each band bin adds $\kappa_{\rm d}(b, \text{leaf}) = \rho\kappa_{\rm leaf}\, s_{\rm abs}(b)$ where $s_{\rm abs}(b) = (1 - a(E_b))C_{\rm ext}(E_b)/C_{\rm ext}(\lambda_{\rm ref})$ comes from a kext table covering the EUV (the WD01/D03 $R_V$=3.1 table, 1 Å–$10^4\,\mu$m: $C_{\rm abs}/{\rm H} = 1.83\times10^{-21}$ at 13.6 eV falling to $3.96\times10^{-22}\,{\rm cm}^2$ at 100 eV). A trap worth recording: the D03 table is *not monotonic* in wavelength (an out-of-order seam) — without sorting, the linear interpolation search silently returned $s_{\rm abs}(100\,{\rm eV}) = 0.903$ instead of 0.832 and shifted the dusty Strömgren radius by +1.6%.
- **Scattering.** With `par%ion_dust_scatter` the scattering coefficient joins the extinction ($s_{\rm sca}(b) = a\,C_{\rm ext}/C_{\rm ext,ref}$; in-band Henyey–Greenstein asymmetry $g(E_b) = 0.67$–$0.99$ from the table $\langle\cos\rangle$ column) and interactions are sampled: after the analytic zero-variance direct walk, a first interaction is forced along the ray with weight $w_1 = 1 - e^{-\tau_{\rm edge}}$, the packet keeps the albedo fraction $a_{\rm eff} = \kappa_{\rm sca,d}/\kappa_{\rm ext}$ at each interaction, scatters into an HG direction, and continues with unforced flights (whose segments tally into $J_b$) under Russian roulette ($w_{\rm min} = 10^{-4}$). Stellar and diffuse packets share the walk; gas ionization, gas heating, and grain heating all derive from the same tally, so the energy split is bookkept by construction.

- **Live ionization-dependent dust with a PAH split.** `dust_model='laursen09_live'` recomputes the dust density from the *computed* $x_{\rm HI}$ every iteration, $n_{\rm d} \propto (Z/Z_{\rm ref})\, n_{\rm H}\left[(1-f_{\rm PAH})(x_{\rm HI}+f_{\rm ion}x_{\rm HII})+f_{\rm PAH}(x_{\rm HI}+f_{\rm ion,PAH}\,x_{\rm HII})\right]$, the PAH share carrying its own ionized-gas survival ($f_{\rm ion,PAH}=0$ default: PAHs destroyed in ionized gas) — the generalization including its PAH-specific option.

- **Grain heating and the equilibrium temperature.** `Heat_dust(leaf)` $= \sum_b [4\pi J\Delta\nu]_b\, \kappa_{\rm abs,d}$ [erg s$^{-1}$ cm$^{-3}$] is consumed by `dust_temp_mod`: $f_{\rm emit}(T) = 4\pi \int s_{\rm abs}(\nu) B_\nu(T)\, d\nu$ is precomputed from the same kext table (log-$T$ grid, 3.2–3200 K) and $T_{\rm d}$(leaf) solves Heat_dust $= (\rho\kappa/d_{\rm cm}) f_{\rm emit}(T_{\rm d})$; outputs are the $T_{\rm d}$ leaf map and the grid-integrated modified-blackbody IR spectrum (1–1000 $\mu$m). FUV grain heating is transported by the `par%add_fuv` band extension (Section 2.1): gas cross sections vanish below their thresholds automatically, so no new transport code is involved, and the two-segment grid keeps the ionizing bins at full resolution. This is the single equilibrium *mixture* temperature; the size- and material-resolved treatment is the SEDust coupling below.

## 4.9 Stochastic dust emission (SEDust coupling)

With `par%dust_sed`, `sedust_mod` (adapted from the MoCafe `dustemis_mod`) computes the grid-integrated dust SED with size-resolved stochastic heating and PAH bands, using the in-tree SEDust library (built from source under `SEDust/sed` by `build_lib.sh`; astrodust, DL07, and Zubko grain models). The coupling follows the MoCafe convention that makes the normalization exact by construction:

- **Shape from the local field.** For each leaf, the band tally $J_b$ is converted to $J_\lambda$ on the SEDust wavelength grid (1129 points, 0.0912–$4\times10^4$ $\mu$m for astrodust). The band reaches into the EUV, below the SEDust optics grid; that energy is deposited into the shortest SEDust bin, conserving the absorbed power (the hardness of the stochastic temperature spikes is slightly underestimated there — a documented approximation). SEDust then returns the emission spectrum of the grain population in this field: equilibrium large grains plus the full temperature-distribution solve for small grains and PAHs.

- **Normalization from the transport.** The SEDust spectrum is used only as a shape: each leaf's emission is normalized to its absorbed power Heat_dust $\cdot\, V$ from the Monte Carlo tally (radiative equilibrium). H II regions are optically thin to the re-emitted IR, so no Lucy re-iteration is needed and the grid-integrated $L_\lambda$ is accumulated directly (an `MPI_ALLREDUCE` over leaves distributed round-robin). $\int L_\lambda\, d\lambda = \sum$ Heat_dust$\,V$ holds by construction and is written into the `_dustsed.txt` header.

- **Cost.** The exact stochastic solve costs $\sim$0.1–0.2 s for each leaf with nonzero grain heating; it runs once, at output time, after convergence (the dusty Strömgren smoke test: $\sim$1.9 hours on 8 ranks for a $64^3$ grid — the solve, not the transport, dominates).

- **Smoke result.** Dusty Strömgren at $4\times10^4$ K: $\int L_\lambda d\lambda = 4.296\times10^{38}$ erg/s $= \sum$ Heat_dust$\,V$ to 4 digits; the SED shows the PAH bands at 3.3/6.2/7.7/8.6/11.3/12.7/17 $\mu$m over a warm 20–60 $\mu$m plateau with the FIR falloff — the canonical stochastically heated spectrum for a hot compact field.

- **Live PAH share** (`par%sed_pah_live`): `dust_emission`'s channel output (astrodust: 'AD' + 'PAH') lets each leaf's spectrum be assembled with the PAH channel weighted by the leaf's PAH survival ($x_{\rm HI}+f_{\rm ion,pah}x_{\rm HII}$), divided under `laursen09_live` by the dust survival already carried by `rhokap`; the Heat_dust $\cdot V$ normalization keeps energy closure (the PAH energy shifts to the large grains). Smoke: the PAH bands drop to 0.26 (7.7 $\mu$m) and 0.15 (11.3 $\mu$m) of the PAH-everywhere reference, the FIR rises by 1.06–1.36×, and the 8-$\mu$m map turns from centrally peaked into a *ring* at the ionization front — the classic Spitzer/JWST bubble morphology.

## 4.10 Peel-off imaging

With `par%ion_peel`, one extra transport pass runs on the converged state with observer peeling (the MoCafe peeling-off convention, adapted to the band): every emitted packet — stellar and diffuse — peels a direct contribution $L_{\rm pkt} w\, e^{-\tau}/(4\pi r^2)$ toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function of the packet's bin (after the albedo weight, before the new direction is sampled). The optical depth to the box edge along the observer direction is integrated through `kap_ion` at the packet's bin by a tally-free walk (tallying the virtual peel ray would double count), so the images carry exactly the transport's extinction: gas, dust absorption, and dust scattering. Images are TAN projections on the MoCafe observer geometry (`obsx/y/z` or angle lists, rotation matrix, auto-sized pixel scale), written as surface brightness [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] split into direct/scattered and ionizing/FUV channels. The band tally is rebuilt by the imaging pass, so the written rates reflect the same field the images carry. With `par%peel_bins` the channel axis becomes the band bin — `direct_cube`/`scatt_cube` $(n_x, n_y, n_\nu)$ — for hardness maps and synthetic EUV/FUV photometry (the band-integrated channel images are still written, derived from the cube; the gate shows the expected color: the FUV scattered halo shrinks from a 15.5-px to a 12.0-px half-light radius between 6.2 and 13.2 eV).

**External-field peel.** The direct peel of an external packet (Section 2.2) replaces the isotropic $1/(4\pi r^2)$ with its Lambert emission density about the entry-surface inward normal $\hat{n}_{\rm in}$, $p(\Omega) = (\hat{k}\cdot\hat{n}_{\rm in})/\pi$, so the contribution is $L_{\rm pkt}\, w\, \max(\hat{k}_{\rm obs}\cdot\hat{n}_{\rm in}, 0)\, e^{-\tau}/(\pi r^2)$: only the far-side surface (inward normal toward the observer) contributes, and the image is the background field seen through the medium — the cloud in silhouette. In the optically thin limit the image-integrated flux is $F = JA_{\rm proj}/d^2$, and with `save_direc0` the ratio `direc0/direct` $= e^{+\tau}$ is again the line-of-sight optical-depth map. The scattered peel is origin-agnostic and unchanged.

For the *dust* bands no transport is needed — the nebula is optically thin to its own IR — so `par%dust_emis_bands` instead stores SEDust band emissivities of each leaf (`_dustemis`, the same block layout as the `_emis` file) and the Python map maker projects them into flux-conserving dust-band images exactly like line maps.

## 4.11 Cartesian (car) grids

`par%grid_type = 'car'` builds a single-level Cartesian grid and stores its leaves in raster order: leaf = cell = $1 + i_x + n_x(i_y + n_y i_z)$. No tree topology, no neighbor table, and no geometry arrays are built. (`'uniform'` is accepted as a backward-compatible alias.) The leaf list comes either from a file (`par%amr_file`, single level) or — with `amr_file` empty — directly from the namelist: `par%nx/ny/nz` cells over the box $[-x_{\max}, x_{\max}]\times\ldots$ (cubic cells required).

**Uniform density and sphere/shell geometry.** A gas-density model is applied on the leaf centers before either grid (`'car'` or `'amr'`) is built: `par%nH_const` overrides every leaf's $n_{\rm H}$ with one uniform value, and `par%rmax/rmin` zero

$n_{\rm H}$ outside a sphere of radius $r_{\rm max}$ / inside $r_{\rm min}$ (a shell). Because the cut is on the leaf centers, a sharp sphere or shell edge can be resolved by the octree (`'amr'` with a refined leaf list) while the density stays exactly uniform — the case that motivates uniform density on `'amr'` as well as `'car'`.

**Analytic geometry.** All geometry queries go through accessor functions in `octree_mod` (`cell_cx/cy/cz/ch`, `leaf_cell`, `leaf_cx/cy/cz/half`). On the octree these are plain array reads; on the car grid they decode the raster index and evaluate $c_x = x_{\rm min} + (i_x + \frac{1}{2})\,\Delta$ — the same floating-point expression that previously filled the arrays, so nothing changes numerically. The car build therefore allocates *none* of the tree arrays: not only `children` ($8\times$), `neighbor` ($6\times$), and `parent`, but also the cell centers, half-widths, and the leaf↔cell maps. At $512^3$ this removes the entire $\sim$10 GB of shared tree memory; the grid costs only its physical fields. All physics modules (thermal, ionization, diffuse, lines, dust) read geometry exclusively through the accessors, so they run unchanged on either grid.

**Traversal (`par%car_walk`).** The car grid is not a physics choice — both walks discretize the same transport integral and agree to rounding — so `par%car_walk` selects only *how* the ray is stepped:

- `'dda'` (default): dedicated incremental Amanatides–Woo walks, the natural car traversal. Each ray precomputes, once, the parametric distance to the next face crossing per axis ($t_{{\rm max},x/y/z}$) and the crossing period ($t_{\Delta,x/y/z} = \Delta/|k_{x/y/z}|$); the inner loop advances by comparisons and additions alone — no exit-face solve at each step, no leaf re-lookup. The tally expressions (analytic first-flight $w(e^{-\tau_{\rm in}} - e^{-\tau_{\rm out}})/\kappa$, pathlength sums, $\tau$-target stop) are identical to the shared walks. It is the fastest mode: $\times 1.64$ over the shared walk, $\times 2.0$ over the octree (total Strömgren run time $13.0 \rightarrow 6.5$ minutes).

- `'shared'` (verification): the octree band walks (`raytrace_ion_to_edge_amr`, `_to_tau_amr`, `_tau_only_amr`) run unchanged; only their two geometry primitives branch on the grid mode — the leaf finder `amr_find_leaf` becomes three integer divisions (`car_index`) and the face-neighbor step `amr_next_leaf` becomes pure integer arithmetic on the raster index (`car_next`). The exit face and distance are still recomputed each step from the cell center and half-width (`amr_cell_exit`), exactly as on the octree. Because the floating-point order along each ray is then unchanged, the same-seed Strömgren run is *bit-identical* to the single-level-octree run ($\max|\Delta x_{\rm HI}| = 0$) — the cross-check the DDA walk cannot provide (it differs at rounding by design, $\max|\Delta x_{\rm HI}| = 2.7 \times 10^{-15}$, $V_{\rm ion}$ identical). This certifies the raster backend rather than serving production (it is $\times 1.25$ over the octree, slower than DDA).

Everything downstream is untouched (the gas modules only consume leaf-indexed arrays) and `refine_front` is octree-only.

## 4.12 Plane-parallel slab illumination

An *xy-periodic slab* models a medium that is horizontally infinite: a packet that leaves an $x$ or $y$ face re-enters at the opposite face, while the $z$ faces escape. The medium may vary only in $z$ (a 1D column, `nx = ny = 1`) or be an arbitrary 3D tile (`nx,ny` $> 1$) repeated periodically. The wrap is source-agnostic — it is a boundary condition on the transport, so an internal source under `xy_periodic` is a horizontally periodic array of sources, and the external slab illumination below rides on the same wrap. In the octree walks the wrap re-enters the packet at the opposite face's exact boundary coordinate (not by $x \pm x_{\rm range}$, which can land a rounding-epsilon outside the box and have the leaf lookup drop the packet); the incremental DDA wraps the integer index (`modulo`), the parametric $t$-coordinates being translation-invariant on a uniform grid.

External illumination (`source_geometry='slab'`) enters the top ($+z$) and/or bottom ($-z$) face, each as a collimated beam at incidence angle $\theta$ or an isotropic (Lambert) field. The physical normalization is the flux crossing the horizontal face,

$$F_z = \int_{\rm hemisphere} I_{\rm in} \cos\theta \, {\rm d}\Omega = \begin{cases} \mu F_n, & \text{beam } (\mu = \cos\theta,\ F_n \text{ normal to the beam}) \\ \pi I, & \text{isotropic } (I \text{ the specific intensity}), \end{cases} \tag{22}$$

so the user inputs the source strength ($F_n$ or $I$) and the geometric factor ($\mu$ or $\pi$) gives $F_z$. The entering luminosity of one periodic tile is $L = F_z A_{\rm zface}$ (the packet weight is $L/N_{\rm phot}$); $A_{\rm zface}$ cancels in every per-volume result, which therefore scales linearly with $F_z$ and is independent of the tile area and lateral resolution. The two faces compose additively (both lit $\Rightarrow L = (F_{z,\rm top} + F_{z,\rm bot})A$).

The emergent observable is the boundary intensity: a packet escaping a $z$-face carries its surviving luminosity into a $(\mu = |k_z|, \text{face})$ histogram, written as $I(\mu)$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] with $F = 2\pi \int I\mu \, {\rm d}\mu$, together with the reflected/transmitted/absorbed energy budget. Both the direct beam (the analytic edge walk) and, with dust scattering, the scattered/diffuse flights contribute, so the budget closes to unity. Point-observer peel-off is disabled under `xy_periodic` (the periodic images admit no finite-distance observer). The quasi-random (`sobol`) launch covers the slab in both modes (a six-dimensional layout: frequency, face, the two in-face coordinates, and the Lambert $\mu, \phi$).

## 4.13 PDR-zone physics

Three switches (default off) extend the FUV zone beyond ion structure:

- `par%metal_ne`: electrons from the metal cascade join the $n_e$ closure of the cell solve, $n_e = n_{\rm H}[x_{\rm HII} + y_{\rm He}(\ldots)] + \sum_{\rm el} A_{\rm el} n_{\rm H} \sum_i (i-1) f_i$ — a $\lesssim 10^{-3}$ correction inside the H II region but the *only* electrons beyond the front ($n_e \simeq n_{\rm CII} \approx A_{\rm C} n_{\rm H}$ in the PDR zone), where they control the [C II]/Mg II excitation and the photoelectric efficiency.

- `par%metal_heat`: metal photoheating enters the thermal balance; the heating integrals $\mathcal{H}_i = \sum_b [4\pi J \Delta\nu]_b \sigma_i (1 - E_{{\rm th},i}/E_b)$ are accumulated together with the metal $\Gamma$'s.

- `par%grain_pe`: grain photoelectric heating and grain recombination cooling with the Bakes & Tielens (1994) fits, $\Gamma_{\rm pe} = 1.3 \times 10^{-24} \epsilon G_0 n_{\rm H}$ with $\epsilon = 4.87 \times 10^{-2}/[1 + 4 \times 10^{-3} x^{0.73}] + 3.65 \times 10^{-2} (T/10^4)^{0.7}/[1 + 2 \times 10^{-4} x]$, $x = G_0 \sqrt{T}/n_e$, and $\Lambda_{\rm rec} = 4.65 \times 10^{-30} T^{0.94} x^{\beta} n_e n_{\rm H}$, $\beta = 0.74/T^{0.068}$, both scaled by `pe_scale` times the leaf's dust amount relative to Milky-Way dust, measured from the leaf opacity itself: $(\rho\kappa/d_{\rm cm})/(n_{\rm H} C_{\rm ext,MW}(V))$ with $C_{\rm ext,MW}(V) = 4.868 \times 10^{-22}\,{\rm cm}^2/{\rm H}$ — robust against the DGR-versus-`cext_dust` input conventions (the first gate run had DGR = 1 with the MW extinction folded into `cext_dust`; a naive DGR/0.01 scale made the heating 100× too strong and drove the zone to a false Ly$\alpha$-capped 12 kK equilibrium). $G_0$ is the local FUV field in Habing units summed from the FUV bins of the tally (`add_fuv` required). Three further guards, each found by the gate: (i) the BT94 fits are evaluated at $\min(T, 10^4\,{\rm K})$ — outside their fit range the $T^{0.7}$ term of $\epsilon$ diverges; (ii) both grain terms carry an $x_{\rm HI}$ weight — the BT94 charging parameter knows only the FUV field, so in ionized gas (EUV-charged grains, grain energy already routed to the IR via `Heat_dust`) it would overheat badly; (iii) the package includes the *H-impact* fine-structure cooling of [C II] 158 $\mu$m and [O I] 63 $\mu$m (two-level, low-density limit; $q_{ul}^{\rm H} = 7.6\times10^{-10}(T/100)^{0.14}$ and $9.2\times10^{-11}(T/100)^{0.67}\,{\rm cm^3\,s^{-1}}$, Barinovs et al. 2005) — the cooling fits are electron-impact only, and with $n_{\rm HI}/n_e \sim 10^3$–$10^4$ in the PDR zone the H collisions ARE the coolant. Without (iii) the zone has no coolant between the fine-structure and Ly$\alpha$ regimes, and the $\epsilon(x)$–$n_e$ feedback (collisional ionization raises $n_e$, neutralizing the grains and raising $\epsilon$ fifty-fold) drives it to a false $\sim10^4$ K equilibrium.

Scope note: this is the atomic PDR skin — no $H_2$/CO chemistry, no line transfer, no dust–gas collisional coupling — so PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

### 4.14 $n$-level solver and line luminosities

`nlevel_mod` evaluates exactly the recipe of Section 3.8 (Clenshaw evaluation of the Chebyshev series) and solves the statistical equilibrium $\dot{n} = 0$, $\sum_i n_i = 1$ (partial-pivot elimination; de-excitation $q_{ul} = 8.629\times10^{-6}\,\Upsilon/(g_u\sqrt{T})$, excitation by detailed balance). `lines_mod` integrates $L_{\rm line} = \sum_{\rm leaf} j\,n_{\rm ion}V$ on the converged state and normalizes to the SH95 H$\beta$. The Fortran chain was verified against the Python reference solver on an identical state: 26 line luminosities agree to four significant digits. With `par%emis_output` the same solve also writes the emissivities leaf by leaf (`_emis`: an `emis_<el>_<stage>` block for each ion and `emis_h` for the SH95 lines, [erg s$^{-1}$ cm$^{-3}$] with $\sum {\rm emis}\cdot V = L_{\rm line}$, verified to $4\times10^{-4}$%), together with `n_H`, $n_e$, $T_e$, the H/He fractions, and the metal stage fractions — a self-contained input for line maps and position–velocity work.

### 4.15 The two-photon continuum: emission, cooling, transport

The metastable $2s$ ($2^1$S in He I) levels decay by simultaneous emission of two photons sharing the transition energy. That one channel enters the code in three separate places, and the three treatments are deliberately *not* the same; they are collected here because the differences are easy to mistake for inconsistencies.

**Emission** (`nebcont_mod`, Section 3.9). The emission coefficient is written per $n_e n_{\rm ion}$ and is fed by the *recombination* cascade, one term for each of the three ions:

| ion | weight | $\nu_0$ [Ryd] | $\alpha_{\rm eff}(2s)$ |
|---|---|---|---|
| H I | $n_e n_{\rm HII}$ | 0.7496 | $0.8368\times10^{-13}(T_4)^{-0.723}$ (Pengelly 1964) |
| He I | $n_e n_{\rm HeII}$ | 1.514 | $6.23\times10^{-14}(T_4)^{-0.827}$ (BSS99) |
| He II | $n_e n_{\rm HeIII}$ | 3.00 | SH95 table, $6.16 \to 2.04\times10^{-13}$ over 5–30 kK |

with the Nussbaumer & Schmutz (1984) shape $A(y)$, $y = \nu/\nu_0$, for H I and (scaled by $Z^6 = 64$ in both $A$ and $A_{2q}$) for He II, and the Almog & Netzer (1989) tabulated shape for He I. Only H I carries the collisional suppression

$$\gamma_{2q} \to \frac{\gamma_{2q}}{1 + n_e q_{2s\text{-}2p}/A_{2q}}, \qquad A_{2q} = 8.2249\ {\rm s^{-1}}, \tag{23}$$

with $q_{2s\text{-}2p} = 5.31\times10^{-4}$ (at $T \le 10^4$ K) falling to $4.71\times10^{-4}\,{\rm cm^3\,s^{-1}}$ (at $T \ge 2\times10^4$ K), so the continuum is quenched above $n_{\rm crit} \simeq 1.6\times10^4\,{\rm cm^{-3}}$. No factor is applied to the helium terms: $A_{2q}$ is $51.3\,{\rm s^{-1}}$ for He I $2^1$S and $526\,{\rm s^{-1}}$ for He II

2$s$, putting their critical densities one to two orders of magnitude higher. The three terms are summed into the total $L_\nu$ of `_nebcont.txt` and also written separately as its last column. Because the population is the recombination cascade alone, collisional excitation of 2$s$ contributes nothing to the emitted continuum, which is therefore a lower bound in the partly neutral gas of an ionization front.

**Cooling** (`cooling_mod`, Section 3.7). The H I line-cooling fit is the $n_e \to 0$ statistical-equilibrium limit of the 25-level CHIANTI atom, and level 2 of that atom *is* 2$s$ (82258.956 cm$^{-1}$, $A = 8.229\,\mathrm{s}^{-1}$, the two-photon rate) with the $1s \to 2s$ collision strength present. Collisional excitation of 2$s$ therefore cools with its full 10.2 eV, and the leading term of the fit, $A_1 = 4.4262\times10^{-17}$ at $T_1 = 1.1844\times10^5$ K, is exactly that excitation temperature.

The cooling must *not* carry the density quenching the continuum does, and does not: proton and electron $l$-mixing converts 2$s$ to 2$p$ and the two-photon continuum into Ly$\alpha$, but the gas loses the same 10.2 eV on either route. (CHIANTI `h_1` tabulates no $2s \leftrightarrow 2p$ collision strength, so the equilibrium solve leaves 2$s$ unquenched — harmless for the same reason. Nor does the level produce a spurious line: `lines_mod` extracts only the H$\alpha$/H$\beta$/H$\gamma$ windows of the collisional component, so nothing is double counted against the continuum.) Helium has no collisional-excitation line cooling at all — `cooling_total` carries recombination, free-free, collisional ionization and the H I line term only — so collisional two-photon cooling of He I $2^1$S and He II 2$s$ is absent, as it is in MOCASSIN, TORUS, Wood et al., and CMACIONIZE.

**Transport** (`diffuse_mod`, Section 4.6). Only helium two-photon photons can ionize anything: the H I continuum stops at 10.2 eV, so omitting it costs nothing. The He I $2^1$S branch is one of the four diffuse channels and carries its H-ionizing fraction 0.56. The He II continuum, which extends to 40.8 eV and can ionize both H and He I, is *not* transported, and neither is He II Ly$\alpha$ — of no consequence wherever He$^{2+}$ is absent (both Lexington benchmarks), an omission to close for the hard planetary-nebula and AGN spectra of Section 6..

# 5. Validation gates and results

## 5.1 Rate integrals vs. analytic attenuation

Uniform sphere ($n_\mathrm{H} = 1$, $R = 1$ pc, $x_\mathrm{HI} = 0.1$, $y_\mathrm{He} = 0.1$), central $10^5$ K source, 16 bins, $10^8$ packets. Against the cell-volume-averaged analytic attenuation with identical bins: $\Gamma_\mathrm{HI}$ mean bias +0.004%, median 0.38% (pure Monte Carlo noise), max 2.6%; $\Gamma_\mathrm{HeI}$ bias +0.011% (Figure 1).

## 5.2 Strömgren sphere

Uniform $n_\mathrm{H} = 100$ sphere, $Q_\mathrm{H} = 10^{49}\,\mathrm{s}^{-1}$ ($T_\star = 4\times10^4$ K, 32 bins), case B, $T_e = 10^4$ K. The reference is an exact 1D radial marching with identical bins and rates — necessarily *Gauss–Seidel* (each shell solved with the already-updated interior opacity): a Jacobi sweep advances the front only one absorption length per pass and never converges. Sharp-edge analytic $R_S = 3.153$ pc; the 1D reference gives $R_\mathrm{eff} = (3V_\mathrm{ion}/4\pi)^{1/3} = 3.0365$ pc (the $\sim$4% gap is physical: He consumes photons and the model re-emits none).

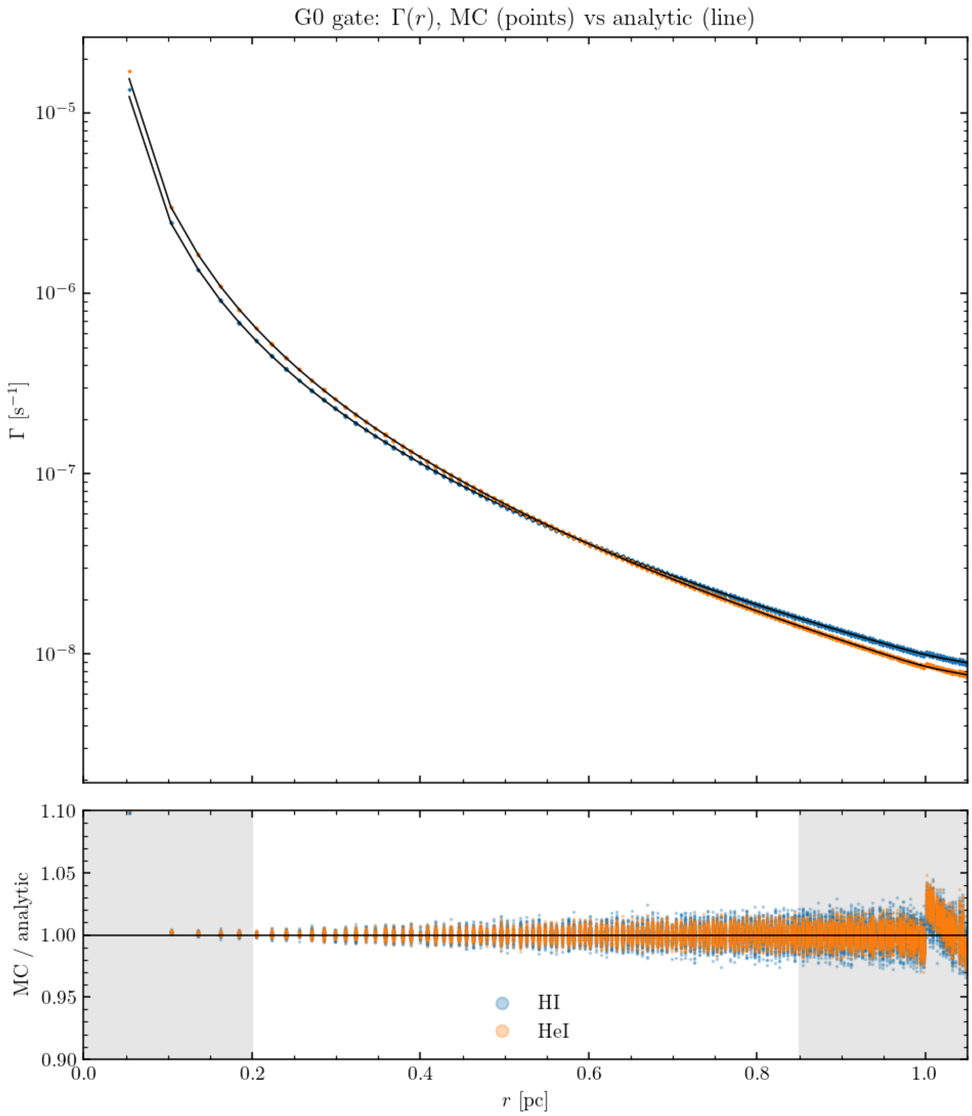


Figure 1: Rate-integral gate: $\Gamma(r)$ from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio (shaded regions excluded: cell-size effects at the center, the voxelized sphere edge outside).

| run | leaves | $R_{\rm eff}$ [pc] | vs. 1D |
|---|---|---|---|
| uniform level 6 | 262 144 | 3.0534 | +0.56% |
| front-refined levels 4–6 | 74 320 | 3.0534 | +0.56% |
| center-refined levels 4–6 | 7 568 | 3.1078 | +2.35% |

The front-refined octree reproduces the uniform grid to four decimals with 3.5× fewer cells; the center-refined control (coarsest cells *at* the front) shows why the I-front is the one place needing resolution (Figure 2).

## 5.3 Thermal H/He nebula

Same configuration with the thermal balance on (Figure 3): $T_e(r)$ matches the 1D thermal reference to median 0.136%, mean bias +0.006%, max 0.95% over $0.3 < r < 2.8$ pc — including the central He III-zone temperature structure and the front drop. Without metals the nebula runs at 19–24 kK, the known pure-H/He overshoot that motivates the metal frame.

## 5.4 The MOCASSIN Lexington benchmarks

HII20/HII40: blackbody 20/40 kK, $Q_{\rm H} = 1.006/4.26 \times 10^{49}$ s$^{-1}$, $n_{\rm H} = 100$, hollow shell $R_{\rm in} = 3\times10^{18}$ cm, abundances He 0.1, C $2.2\times10^{-4}$, N $4\times10^{-5}$, O $3.3\times10^{-4}$, Ne $5\times10^{-5}$, S $9\times10^{-6}$. The stored MOCASSIN 2.02.73.2 baselines are *reduced* runs ($10^5$ photons, 3 iterations) — at iteration 3 only 27% of cells have converged. To close the comparison MOCASSIN itself was re-run converged ($2\times10^6$ photons, 40 iterations, 72% cells; a finer nx=25 grid confirms resolution), and MoCHII with the *corrected* charge exchange of Section 3.6.

**Temperatures.** $T_e$ weighted by $n_{\rm HII} n_e$, MoCHII with its current defaults (threshold-aligned grid, corrected charge exchange of Section 3.6, and the density-suppressed cooling of Section 3.7); the benchmark inputs carry the ionizing-band luminosity that reproduces the specified $Q_{\rm H}$, transport the diffuse field, and use case-B recombination lines:

| | MoCHII (`local_ne`) | MoCHII (`low_density`) | CLOUDY c25.00 | Lexington-2000 (Ercolano+03) | MOCASSIN 40-it |
|---|---|---|---|---|---|
| HII20 | 7025 K | 6376 K | 6910 K | 6402–6749 K | — |
| HII40 | 8211 K | 7894 K | 8210 K | 7720–8199 K | 8176 K |

With the density-suppressed cooling default (`par%cooling_model = local_ne`) MoCHII gives $T_e = 8211$ K (HII40) and 7025 K (HII20), matching a current CLOUDY c25.00 run (8210 and 6910 K) to 0.7 K and 1.7%. Both modern codes sit a few hundred K *above* the 2000-era published ranges of Ercolano et al. (2003) — $\langle T[N_p N_e]\rangle = 7720$–8199 K (median 8026) at HII40 and 6402–6749 K (median 6662) at HII20 — whose tabulated columns carry older atomic data; the two modern codes now agree with each other far more closely than either does with the historical spread.

The ∼9% HII40 excess reported in earlier revisions (a superseded $T_e = 8913$ K) is *resolved*: it was two defects in the Tier-1 cooling fits (Section 3.7), compounded by the then-missing density suppression. **(D1)** The ground-term fine-structure populations were distributed by Boltzmann factors — the *high*-density limit — instead of the $n_e \to 0$ limit, which suppressed the far-infrared fine-structure cooling ([O III] 52/88 $\mu$m, [N III] 57 $\mu$m, [C II] 158 $\mu$m) by factors 1.6–2.9. **(D2)** The `.scups` Burgess–Tully *scaling* energy was used as the physical transition energy in both the

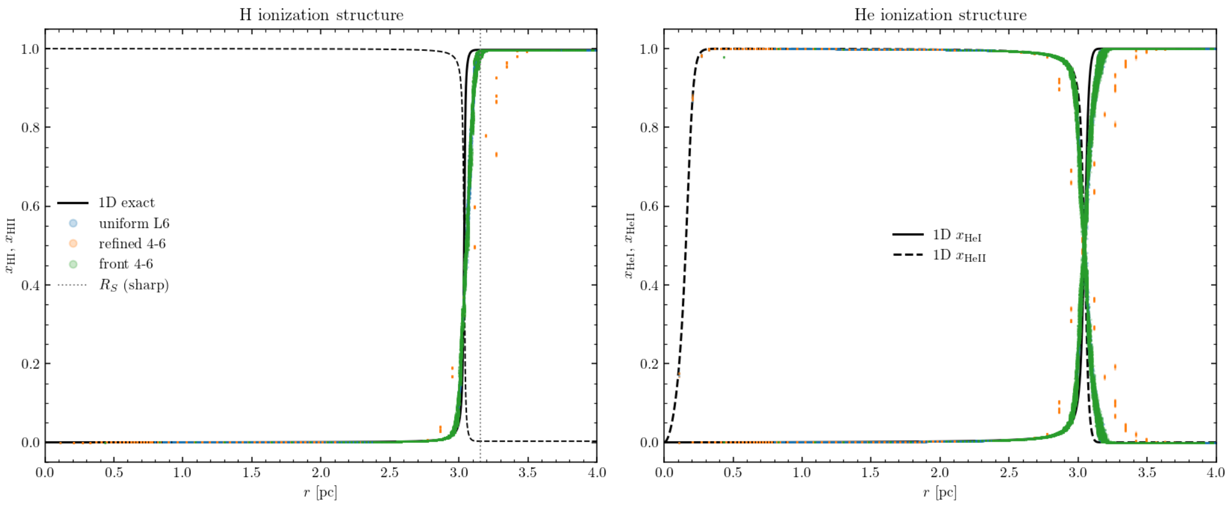


Figure 2: Strömgren gate: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (orange), front-refined (green, overlying blue).

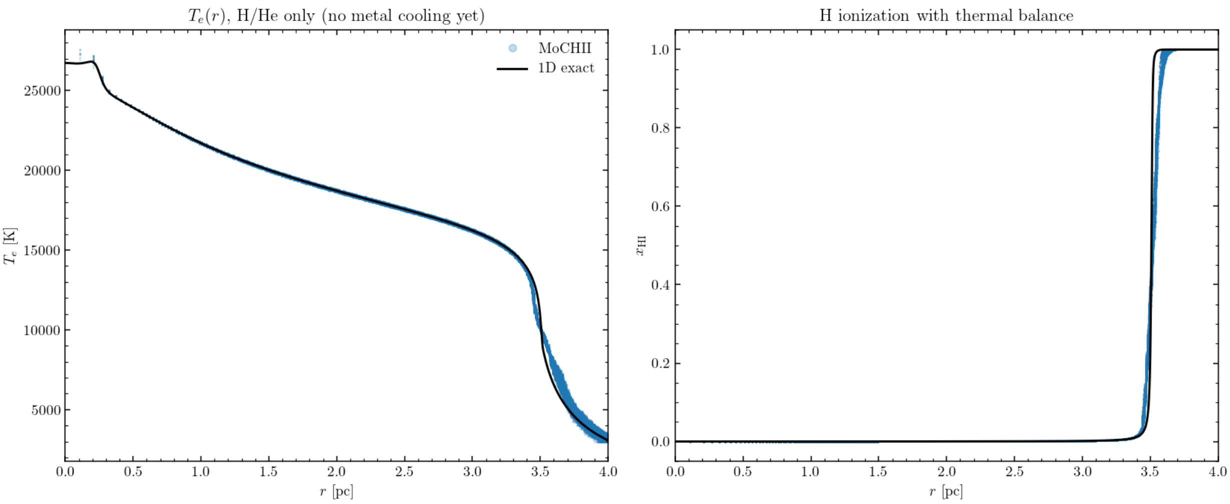


Figure 3: Thermal gate: $T_e(r)$ and $x_{\rm HI}(r)$ with the thermal balance, against the 1D exact reference.

Boltzmann factor and the radiated photon, in place of the observed `.elvlc` level difference (e.g. [O II] cooling low by 0.58× at 8 kK). Both were located by a fixed-state cooling-budget comparison against CLOUDY c25.00 (its `save cooling each` output): D1 was 23.9% and D2 10.6% of the HII40 cooling budget, while missing channels ([S IV] 10.5 $\mu$m at 1.2%, now added; all others < 0.2%) and trace-species coupling ($n_e$ plus metal photoheating, together +0.2%) were excluded. Regenerating the fits from the $n_e \to 0$ statistical-equilibrium limit with observed energies gives HII40 7894 K and HII20 6376 K (the `low_density` option); restoring the density suppression at the local $n_e$ (the `local_ne` default) raises them to 8211 and 7025 K. The former "cooling-side residual" was thus the low-density over-cooling of the low-$n_{\rm crit}$ fine-structure lines, and there is no open temperature discrepancy with the current reference code.

**Ionization structure** ($n_e$-weighted means, HII40). The He and metal ionization is set correctly once the ionizing grid resolves the He I edge. With `par%ion_align_edges` on (the default) or nnu_ion ≥ 128 the He-ionizing photon budget returns to the exact blackbody value ($Q_{\rm HeI}/Q_{\rm HI}$ = 0.108 against 0.127 for a 32-bin log grid), He I rises from 0.058 to 0.163 (MOCASSIN 0.207, closing most of the gap), and the metal 3+ fractions fall correspondingly. A separate in-code swap of MOCASSIN-style recombination (Pequignot 1991 RR + Nussbaumer–Storey 1983 DR) showed the recombination-coefficient vintage to be a minor lever: MoCHII's Badnell 2023 $\alpha$(X III → X II) runs ∼0.8× MOCASSIN's, but the swap moved the fractions by only ∼0.026 and closed ∼17% of the ionization gap. The table below is the old 32-bin log grid, the reference point for the alignment discussion (corrected CX vs. the converged 40-iteration MOCASSIN):

| ion | MoCHII (log) | MOCASSIN | ion | MoCHII (log) | MOCASSIN |
|---|---|---|---|---|---|
| C II | 0.188 | 0.355 | O II | 0.344 | 0.491 |
| C III | 0.805 | 0.640 | O III | 0.653 | 0.496 |
| N II | 0.180 | 0.386 | Ne II | 0.524 | 0.587 |
| N III | 0.818 | 0.600 | Ne III | 0.474 | 0.401 |
| S II | 0.194 | 0.066 | S III | 0.750 | 0.786 |

Switching on the aligned-grid default brings the He and metal fractions substantially closer to MOCASSIN — He I 0.058 → 0.163, O III 0.653 → 0.601, C III 0.805 → 0.694 — so most of the apparent over-ionization in the table is the log-grid discretization at the He I edge, not a physics error. At HII20 the CX correction is negligible (low excitation): the structure matches everywhere except Ne I/II (the 21.6-eV threshold at a 20-kK source, a known code-spread hotspot) and the S II/III split.

**Line fluxes** (ratios to H$\beta$). With the corrected cooling the optical collisional lines — formerly systematically high because $T_e$ ran ∼9% hot — now fall inside the published spread: [O III] 5007 drops from ∼4.3 to ∼2 against the published 1.64–3.27, and the [N II], [S II] and [Ne III] ratios move into range. Absolute $L$(H$\beta$) ≈ $1.9 \times 10^{37}$ (HII40) and $4.7 \times 10^{36}$ erg s$^{-1}$ (HII20) sit ∼3% below the published 2.01–2.10 × $10^{37}$ and 4.83–5.04 × $10^{36}$, consistent with helium absorption of the >24.6-eV photons and the weak temperature dependence of $\alpha_{\rm B}$; with the corrected cooling the HII40 front is captured within the prescribed sphere (outermost-shell $x_{\rm HII}$ = 0.11, recombinations 95% of $Q_{\rm H}$). The emissivity chain itself is exact (Section 4.14).

## 5.5 Re-refinement gate and the TNG demonstration

The re-refinement of Section 4.7 on the Strömgren configuration (level-5 start, rebuild to level 7 at iteration 15, 32 768 $\to$ 464 248 leaves):

| run | leaves | $R_{\mathrm{eff}}$ [pc] | vs. 1D |
|---|---|---|---|
| re-refined 5$\to$7 | 464 248 | 3.0432 | +0.22% |
| native uniform level 7 | 2 097 152 | 3.0430 | +0.21% |

The two agree to +0.008% with 4.5$\times$ fewer leaves, and both improve on level 6 (+0.56%) — resolution-convergent, as the resolution-convergence requirement expects. The front profiles overlie exactly.

The post-processing demonstration runs MoCHII on an Illustris-TNG 60-kpc cutout (the MoCafe converter output, $n_{\mathrm{H}}$ and $x_{\mathrm{HI}}$ columns from the simulation) with a $Q_{\mathrm{H}} = 10^{53}\,\mathrm{s}^{-1}$, 40-kK source at the density peak, thermal balance + metals + re-refinement (Figure 4): dense clumps cast sharp neutral shadows, the circumgalactic gas sits at $x_{\mathrm{HI}} \sim 10^{-4}$ with $n_e$-weighted mean $T_e = 7502$ K, and the refinement (32 768 $\to$ 617 135 leaves) tracks the *distributed* ionization topology of a turbulent medium — clump skins and partially ionized gas, not a single thin shell. The run also demonstrates why the max-cell convergence test stalls in such media (clump-surface jitter); a volume-integrated criterion is queued.

## 5.6 Dusty Strömgren gate

With the MW D03 dust of Section 4.8 added to the Strömgren configuration ($\tau_{\mathrm{d,abs}}(13.6\,\mathrm{eV}) = 1.7$ over the dust-free radius), against the 1D reference carrying the identical dust absorption in every bin:

| case | $R_{\mathrm{eff}}$(1D) [pc] | $R_{\mathrm{eff}}$(MC) [pc] | dev |
|---|---|---|---|
| dust-free | 3.0365 | 3.0534 | +0.56% |
| full MW dust | 2.1983 | 2.2058 | +0.34% |
| `laursen09_live` | 3.0218 | 3.0425 | +0.68% |

The classic dusty reduction ($R/R_0 = 0.724$ in 1D, 0.722 between the two MC runs) is reproduced within the gate tolerance; the live ionization-dependent dust drives itself to near dust-free ($R/R_0 = 0.996$: interior dust vanishes with the computed $x_{\mathrm{HII}}$), confirming the dust–ionization feedback loop converges self-consistently. Energy budget: dust absorbs 61% of the band luminosity in the full-dust case ($L_{\mathrm{abs}} = 1.933 \times 10^{38}$ of $3.178 \times 10^{38}\,\mathrm{erg\,s^{-1}}$).

**Scattering bracket.** Turning the scattering on must land the sphere between two analytically defined bounds: treating the dust extinction entirely as absorption (scattered photons destroyed; 1D $R_{\mathrm{eff}} = 2.059$ pc) and the absorption-only treatment (scattered photons never interact; 2.198 pc, MC counterpart 2.206 pc). The full-scattering run gives $R_{\mathrm{eff}} = 2.197$ pc — inside the bracket and close to the absorption-only bound, exactly as the strong forward scattering ($g = 0.67$–$0.99$) predicts: forward-scattered photons continue nearly radially, and only the small diffused fraction is lost from the radial budget. The dust-absorbed luminosity rises by 0.7% (rescattered photons reabsorbed) — energy bookkeeping consistent.

**Dust temperature and IR.** With the band widened to 6 eV (the band luminosity rescaled by the exact blackbody ratio 2.0969 so $Q_{\mathrm{H}}$ is preserved) and scattering on, the equilibrium mixture temperature runs from $T_{\mathrm{d}} = 87$ K at 0.3 pc to 28 K at the sphere edge — the canonical HII-region gradient — and the IR spectrum integral closes on the absorbed power to

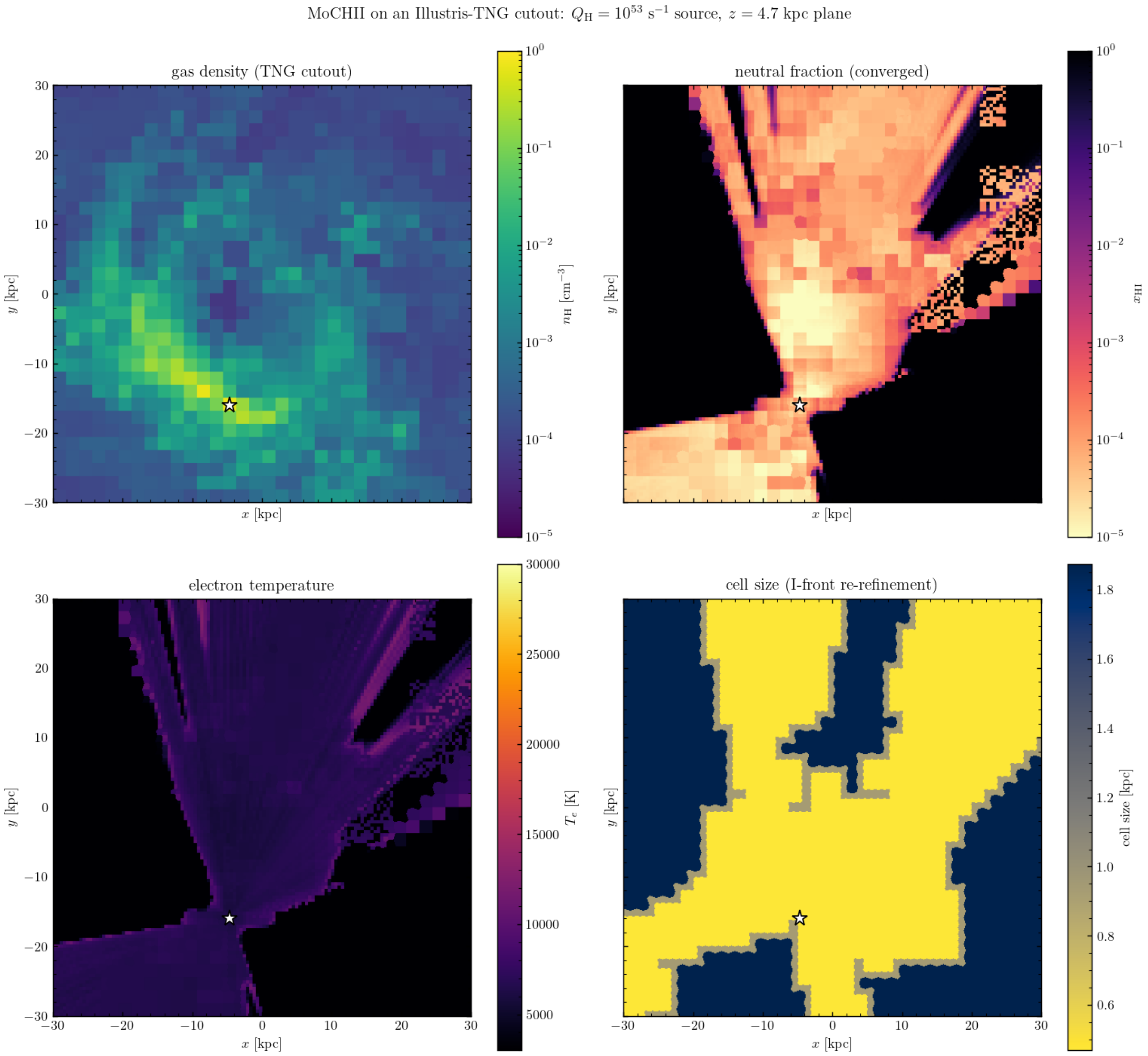


Figure 4: MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

$L_{\rm IR}/L_{\rm abs}$ = 1.000. (This demo predates the two-segment `add_fuv` grid and rescaled the single-segment band by hand; the +1.5% bin-coarsening shift it measured is what the two-segment grid now removes.)

### 5.7 FUV extension gate: PDR-side C II and Mg II

The `add_fuv` smoke test (dusty Strömgren, 32 ionizing + 16 FUV bins, thermal + metals with Mg/Fe, dust scattering on; `tests/d_dusty/d_fuv_opt.in`) verifies the band mechanics and the physics it was built for. Mechanics: the two-segment grid reports $L_{\rm FUV}/L_{\rm ion}$ = 1.0969 for the $4\times10^4$ K Planck source at a 6 eV edge — exactly the blackbody band ratio applied by hand in the earlier demo — with $Q_{\rm H}$ preserved by construction. Physics: inside the front (2.2 pc with this dust and cooling) the metals sit at C III/Mg III as before; *beyond* it, where H is neutral and the ionizing bins are exhausted, the FUV bins propagate under dust extinction alone and hold the low-threshold stages fully singly ionized — radially averaged C II = 1.000 and Mg II = 1.000 from the front out to the sphere edge, the PDR-side structure that the ionizing-only band cannot produce (there the same zone recombines to C I/Mg I). Two documented limitations in that zone: electrons from metal ionization are not fed back into $n_e$ (the trace-species design), and with photoheating $\approx 0$ the thermal bisection pins $T_e$ at `te_min` — PDR temperatures require photoelectric heating, outside the present scope. The front-zone cells with vanishing heating also dominate the max $|\Delta T_e|$ convergence test (another argument for the queued volume-integrated criterion); the H II-interior structure and $T_e$ (8.1–10.5 kK) converge normally.

### 5.8 External field and multiple sources

The source-component model of Section 2.2 is gated in the `ext_field`, `multi_src`, and `peel_ext` directories under `tests/`:

- **External field.** In an optically thin box ($\tau \sim 7\times10^{-3}$) the interior band-integrated mean intensity recovers $J_{\rm ext}$ to 0.9981 (`'rec'`) and 0.9991 (`'sph'`), uniform and isotropic (axis asymmetry $<0.1\%$); the entering power equals $\pi J A_{\rm surf}$ [Eq. (1)] exactly.
- **Superposition.** Two point sources against the analytic sum $\sum_i L_i/(16\pi^2 r_i^2)$: median deviation 0.76%, mean ratio 0.9989; a mixed point + external run, median 0.75%. Two-temperature additivity — a two-source run against the sum of the two single-source runs — agrees to a bin-wise median of 0.12% and to 1.0001 in the band total.
- **Spectra.** A multi-column `src_spectrum_file` tabulating the same Planck curves as its `src_tstar` twin reproduces it to 0.010% in the worst bin; `ext_spectrum` against its `ext_tstar` twin to 0.019% (this check also caught and fixed the single-component path ignoring `ext_tstar/ext_spectrum`).
- **External peel.** The thin-box image-integrated direct flux against $F = JA_{\rm proj}/d^2$: –0.22% (`'rec'`); the `direc0` flux +0.15% at chord $\tau \sim 2$; the measured central-pixel $\tau_b = -\ln(\mathrm{direct}/\mathrm{direc0})$ reproduces $n_{\rm H}\,\sigma_{\rm HI}(E_b)\,\ell$ bin by bin to four digits with the exact hydrogenic cross section (deviating from the $E^{-3}$ power law by up to $\sim$10%, as $\sigma_{\rm exact}$ does).
- **Physical-unit spectra.** A `'per_ev'` source against its `'shape'` twin agrees to 0.004% in the worst bin (the derived `src_lum` is exact); the `'per_ang'` and `'per_um'` wavelength conversions reproduce it to 0.013%; rescaling a

physical file to a set scale matches its shape twin to 0.0000%. A physical `ext_spectrum` $J_E$ file recovers the interior $\langle J\rangle/J = 0.998$, and doubling `ext_intensity` doubles the field exactly (2.0000).

- **ISRF presets.** The measured FUV energy density matches the analytic value to $u_{\rm FUV,meas}/u_{\rm FUV} = 1.0002$–$1.0003$: 8.94 (Draine), 5.23 (Habing), $6.01\times10^{-14}$ erg cm$^{-3}$ (Mathis), with exactly zero in the ionizing bins.
- **Backward compatibility.** Single-source runs are bit-compatible with the pre-change binary (worst difference 1 ULP, $9.9\times10^{-16}$ relative).

## 5.9 Plane-parallel slab

The slab of Section 4.12 is gated under `tests/slab/` against the analytic plane-parallel transport, with the `xy_periodic` wrap exercised throughout:

- **xy-periodic wrap.** A tall thin periodic column ionizes fully ($\langle x_{\rm HII}\rangle = 0.9999$) where the isolated (non-periodic) control reaches only 0.7072 — the wrap re-injects the lateral flux; the octree (`car_walk='shared'` / `grid_type='amr'`) and the DDA agree to $8.5\times10^{-17}$ for `nx`$\geq$2, and after the exact-face wrap fix (Section 4.12) also at `nx=1` ($1.3\times10^{-18}$). The AMR leaf-list and car density-cube file inputs both run correctly under `xy_periodic`.
- **Beam attenuation.** A frozen neutral slab illuminated by a collimated beam gives a photoionization-rate profile $\Gamma_{\rm HI}(z)$ that is a *perfect* single exponential ($R^2 = 1.00000$); at $\theta = 60^\circ$ ($\mu = 0.5$) the decay slope is exactly slope$_0/\mu$ (ratio 1.0000) — the $e^{-\tau/\mu}$ law and the $\cos\theta$ convention are correct. The isotropic mode gives $F_z = \pi I$ exactly and a shallower monotone profile.
- **Emergent $I(\mu)$ and energy budget.** A normal beam transmits $e^{-\tau_\perp} = 0.8874$ exactly (budget closes, $0.88739 + 0.11261 = 1.0$), with $I(\mu)$ a single spike at $\mu = 1$ and zero reflection (no scattering); a $\theta = 60^\circ$ beam transmits $e^{-\tau/0.5} = 0.7874$ exactly; isotropic illumination gives a broad $I(\mu)$ over all bins (mean $\mu = 0.583$, forward-shifted by the $e^{-\tau/\mu}$ weighting). Two-face illumination is a symmetric superposition.
- **Photoionization slab.** A full ionization solve (case B OTS) builds a plane-parallel ionization front at the Strömgren depth $d = \Phi/(n_{\rm H}^2\alpha_B)$; the front position matches to within $\sim$5–7% (recombination-coefficient and finite-front-width uncertainty) and scales with the incident flux ($d \propto F_z$, ratio 1.94 for a factor-of-two change, the small deficit being the roughly constant front width).

## 5.10 Argon as a data operation

Adding argon touched no algorithm: two cooling fits (Ar III, Ar IV), two $n$-level files, one element file (VFKY96 rows $Z$=18, Voronov, Badnell RR + SVS82 DR2, KF96 CX), one abundance parameter. PyNeb cross-check: [Ar III] 7136/7751 within 0.7%; [Ar IV] 4740/4711 within 4.0% (a density pair — the same database-spread class as [O II] and [S II]; PyNeb uses RB97/RGJ19 there). End-to-end smoke test: [Ar III] 8.99 $\mu$m, 7136, 7751 appear in the line table with 7136/7751 = 4.17, the $A$-ratio, as expected.

## 5.11 Magnesium and iron

Mg and Fe repeat the recipe with three new data wrinkles, all absorbed by the readers: the Fe II power-law RR (`RR2`), the SVS82-form DR (`DR2`), and the Mg/Fe charge-exchange forms and stage attribution (Section 3.6). Products: cooling fits Mg II (0.05% max error), Fe II (0.11%), Fe III (0.42%); $n$-level files Mg II (3 levels), Fe II (13 levels, the IR/optical forbidden lines), Fe III (14 levels); element files with VFKY96 $Z$=12, 26 rows. PyNeb cross-check: the [Fe III] 4658/4702 ratio agrees to −6% to −16% — the largest gate deviation so far, and a known property of iron: PyNeb's default Fe III collision strengths (BB14) differ from CHIANTI v11's by tens of percent (the iron collision-strength spread in the literature), compounded by the 14-level truncation of a 34-level model. The number is recorded as a data-vintage bracket, not a code error — the same chain reproduces [O III] to 0.6%. For iron-line studies `par%fe_levels_full` loads expanded 16/34-level Fe II/III models (`nlevel_fe_{2,3}_full.txt`, 298 [Fe III] lines vs 38, e.g. 4658/4702 +3%) in place of the compact defaults. End-to-end smoke test (HII20 sphere with $n_{\rm Mg}/n_{\rm H} = 3\times10^{-6}$, $n_{\rm Fe}/n_{\rm H} = 3\times10^{-7}$, depleted gas-phase values): Mg loads 3 stages, Fe 4 stages; the line table gains the Mg II 2796/2803 doublet (ratio 1.98, the $A$-ratio limit) and the [Fe II] 1.257 $\mu$m, 5.34 $\mu$m, and 26 $\mu$m lines.

## 5.12 Silicon, chlorine, calcium (five stages)

Si, Cl, and Ca bring the registry to eleven elements, each tracked through five stages. The one framework extension was chlorine's cross section (the VY95 path of Section 3.1 — Cl is absent from the VFKY96 outer-shell table); Si carries the KF96 charge exchange from the MOCASSIN `chex` rows (recombination direction), Cl and Ca have no transcription available. $n$-level files (all $\leq$ 0.65% worst transition): Si II/III/IV, Cl II/III/IV, Ca II, Ca V; Cl V and Ca III/IV have no CHIANTI v11 level data and are tracked without lines (the Ar II pattern). Cooling fits: Si II 1.1%, Si IV 0.2%, Ca II 0.4%; Si III 12% and the Cl ions 8–9% resist the multi-exponential form — accepted, since at $\sim3\times10^{-7}$ abundances their share of the thermal balance is $\sim10^{-4}$ and the diagnostics run through the exact $n$-level solve. PyNeb gates: [Cl II] 8579/9124 to 1.5%; Si III] 1892/1883 to 6.9% and [Cl III] 5518/5538 to 6.6% (PyNeb's DK94/BZ89 compilations against CHIANTI v11 — the established database-spread class). End-to-end smoke: [Si II] 34.8 $\mu$m enters the line table at 4.4% of H$\beta$ (the strongest metal line after [C II] — the ionization-front/PDR coolant it is), with Si III] 1883/1892, Si IV 1394/1403, [Cl II] 8581/9126, and the [Cl III] pair at 5518/5538 = 1.402, the low-density limit.

## 5.13 Line and continuum outputs

The converged-state outputs now include: the SH95 H I recombination lines (smoke-state check: H$\alpha$/H$\beta$ = 2.94 at $\sim$7 kK); the $n$-level collisional lines for the 24 registry ions (including [C II] 158 $\mu$m, C III] 1909, [N III] 57 $\mu$m, Mg II 2796/2803, and the [Fe II] IR lines); and the grid-integrated nebular continuum $L_\nu$ (free-bound + free-free from the verified tables and the Milne extension above the edges, plus the three two-photon continua), written on a 400-point log-$\nu$ grid from 11.4 $\mu$m to 217 Å.

## 5.14 He I $2^3$S metastable and 10833 Å absorption

The $2^3$S diagnostic (Section 3.11) is gated in `tests/hei_meta/` on three runs (low/high density and an `add_fuv` run). **Stage 1**: the data-file analytic limits reproduce the low-density ratio $\alpha_3 n_e/A_{31} = 1.6509\times10^{-7}$ (spec $1.65\times10^{-7}$ at

$n_e = 100$, $10^4$ K), the plateau $6.513\times10^{-6}$, and $n_{\rm crit} = 3945$ cm$^{-3}$ (the $n_e$-scan half-plateau crossing lands at 3926); the Fortran module $n_3$ matches the independent Python closed form on each run's own state to worst relative $4\times10^{-15}$ (machine precision); the low-density core sits within median 2.8% / max 3.9% of the radiative limit and the high-density run ($n_e = 1.1\times10^4 > n_{\rm crit}$) at 0.676 of the plateau; $\Phi_3$ from the band tally matches an independent $J_\nu$ integration to $1.5\times10^{-9}$ (max $9.07\times10^{-5}$ s$^{-1}$); and $\alpha_3(10^4\,{\rm K}) = 0.998 \times 0.75\,\alpha_B({\rm He\,II})$, the diffuse `HEI_F3S` consistency. **Stage 2 (10833 absorption)**: the LINE10833 constants carry the $f$ ratio 1:3:5 and $A = 1.0216\times10^7$ s$^{-1}$ each (NIST/p-winds, not the CHIANTI $1.15\times10^7$); a synthetic uniform cube reproduces the column conservation $\sum NA_{\rm pix} = \sum nV$ and the interior column $n_3\,\ell$ exactly, and the line-center $\tau_0 = \int\kappa_0\,{\rm d}l$ per component and for the doublet matches the hand integral $\kappa_0^{\rm hand}\,\ell$ to $< 2\times10^{-16}$; the optically-thin case raises no warning while a $\times10^5$ scaled case pushes $\tau_0 > 1$ and fires it. On the high-density run the column conserves to $9\times10^{-15}$ and the doublet/$J'$=0 $\tau_0$ ratio is 8.003, equal to the $(f_1{+}f_2)/f_0 = 8.002$ oscillator-strength ratio of the blended lines (that run's metastable column is optically thick, $\tau_0 \sim 10^4$ — flagged by the warning, as designed).

# 6. Known limitations and next steps

- By default He recombination emits only the three ground continua; the He I excited-channel ionizing photons ($\sim$60% of He recombinations) are re-emitted only when `par%hei_diffuse` is set — their effect on the He I/II split against MOCASSIN is small (the split is set by the >24.6-eV budget). The He II excited cascade has no such channel: its Ly$\alpha$ (40.8 eV) and two-photon continuum can ionize H and He I but are not transported (Section 4.15). Immaterial wherever $He^{2+}$ is absent, as in both Lexington benchmarks; it belongs to the hard-spectrum track.
- The EUV part of the leaf field, below the SEDust optics grid, is folded into the shortest SEDust bin (energy-conserving; spike hardness slightly underestimated).
- The thermal-loop line cooling is suppressed to the local $n_e$ by default (`cooling_model = local_ne`) through the $(T, n_e)$ table of Section 3.7, and the H-impact [C II] 158 $\mu$m / [O I] 63 $\mu$m term (`metal_cooling_H`, weighted by $x_{\rm HI}$) carries its own two-level saturation. What remains is the *cold* limit: the cooling fits hold over $10^3$–$10^5$ K and collapse below that floor (their lowest excitation temperature is 2054 K for C I, far above the 23.6 K of [C I] 609.1 $\mu$m), and no H-collider term exists for the [C I] fine-structure lines — so gas colder than $\sim10^3$ K, where hydrogen is the collider, is outside the cooling model. Ar II carries no cooling or lines (no CHIANTI v11 level data).
- PDR zones: with the Section 4.13 switches off (the default), the FUV extension gives the PDR-side *ion structure* only; with them on, $n_e$, metal photoheating, and photoelectric heating make the PDR temperatures first-order estimates — $H_2$/CO chemistry and line-transfer cooling remain out of scope.
- The cell-max convergence test stalls at the front-cell Monte Carlo noise floor (and at clump-surface jitter in turbulent media); the volume-integrated criterion (`conv_crit = 'vol'`) resolves this and is the recommended production setting.

- The MOCASSIN Lexington benchmark temperatures are resolved (Section 5.4): with the density-suppressed cooling default MoCHII (HII40 8211 K, HII20 7025 K) matches a current CLOUDY c25.00 run to 0.7 K and 1.7%, the earlier excess having been traced to two Tier-1 cooling defects (D1/D2) and the missing density suppression, all fixed. The one remaining note is that both modern codes sit a few hundred K above the 2000-era published Lexington temperatures — a data-vintage difference, not a code defect.
- Further metal ions as science needs them.